

Monetary Policy under Uncertain Expectations in an Emerging Economy^{*}

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Abstract

Monetary policy reforms in emerging market economies are often shaped by histories of high inflation, which give rise to persistent concerns about the anchoring of private-sector expectations. Against this background, we embed near-rational expectations (NRE) into a small open economy New Keynesian model and characterize robustly optimal monetary policy when the central bank faces Knightian uncertainty about private-sector belief distortions. Applying the framework to Mexico's post-Peso Crisis monetary policy reforms during 1998–2007, we estimate both NRE and rational-expectations (RE) versions of the model. The estimated NRE model implies a substantial concern for robustness and decisively outperforms RE alternatives in tracking the policy rate, which is not used in the estimation. The robust policy rule alone does not deliver this: with the same rule but rational private agents, the model fails on the policy rate. The identification is an open-economy phenomenon: re-estimated as a closed economy, the NRE model collapses to its RE counterpart. The concern for robustness leads the central bank to stabilize domestic inflation more aggressively while tolerating larger nominal exchange rate adjustments, and it generates endogenous history dependence, producing persistent inflation dynamics even in the absence of backward-looking price setting.

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1 Introduction

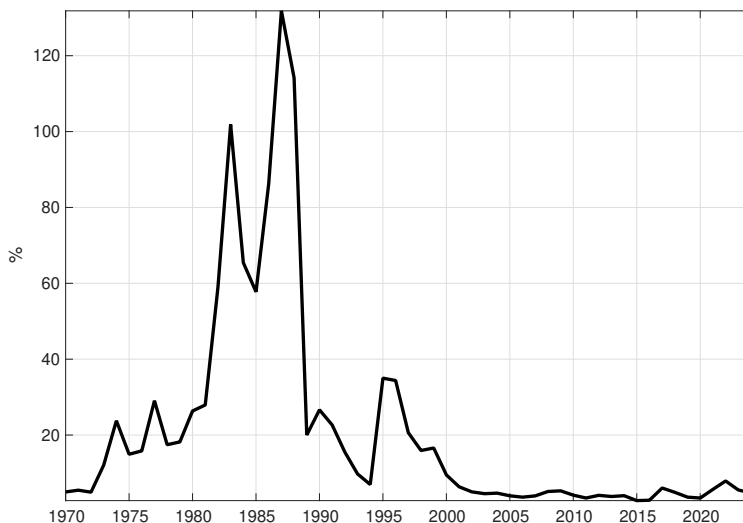
Stabilizing inflation is a core responsibility of central banks worldwide, with the anchoring of inflation expectations playing a central role in this objective. Emerging market economies (EMEs), however, often operate in economic environments that differ markedly from those of advanced economies (AEs), reflecting histories of high inflation, macroeconomic instability, and weaker institutional credibility. Against this backdrop, monetary policy regime shifts in EMEs tend to occur in the aftermath of crisis episodes—such as in Mexico (1994), Thailand (1997), Russia (1998), Brazil (1999), and Argentina (2002)—that compel policymakers to reform their monetary frameworks in an effort to restore macroeconomic stability.

One of the central challenges faced by policymakers during such transitions is how to anchor private-sector inflation expectations. In these reform episodes, the post-crisis policy regime represents a sharp break from the past. Even when a central bank adopts an inflation-stabilizing monetary framework similar to those in AEs, it may lack full confidence about the nature of the expectations it needs to stabilize. In particular, a history of high inflation raises concerns that private-sector beliefs may not be fully aligned with full-information rational expectations (RE), even after credibility-enhancing reforms are implemented. Consistent with this concern, [Jácome et al. \(2025\)](#) document that inflation-targeting central banks in countries with high-inflation histories respond more aggressively to deviations of inflation expectations from the target, even as institutional credibility improves over time. As a result, a persistent tension can arise between institutional reform and inherited expectations, suggesting that concerns about expectation anchoring continue to shape monetary policy responses in post-crisis environments.

In this paper, we develop a framework to study the conduct of monetary policy during periods of monetary regime transition, when central banks face challenges in stabilizing inflation and inflation expectations. Specifically, we analyze the uncertainty surrounding the anchoring of inflation expectations faced by central banks through the lens of Knightian uncertainty, or ambiguity ([Gilboa and Schmeidler, 1989](#)). We study an optimal policy problem in a small open economy New Keynesian (SOE-NK) framework, which allows monetary policy to operate in an environment where external conditions and exchange rate movements interact with domestic inflation dynamics. The policymaker is uncertain about the probabilistic nature of private agents’ belief formation. While private agents may form near-rational expectations (NRE) centered around the rational-expectations (RE) benchmark, the central bank faces ambiguity regarding the private-sector belief distortions. In this environment, the policymaker exhibits ambiguity aversion and seeks a robustly optimal monetary policy that minimizes losses under worst-case expectations, following the approach of [Woodford \(2010\)](#). The remainder of the model environment follows the standard SOE-NK setup ([Galí and Monacelli, 2005](#)) and incorporates a lagged domestic inflation term and a cost-push shock in the aggregate supply relation.

We apply this framework to an analysis of Mexico’s monetary policy reforms following the 1994

Figure 1: Annual Inflation Rate, Mexico: 1970-2024



Source: International Financial Statistics database, International Monetary Fund. Indicator code: FP.CPI.TOTL.ZG.

Peso Crisis, treating Mexico as a canonical case study of post-crisis monetary regime transition in EMEs. As illustrated in Figure 1, Mexico experienced prolonged episodes of high and volatile inflation from the 1970s through the mid-1990s, reflecting chronic macroeconomic instability before the crisis. The Peso Crisis made clear that these structural vulnerabilities had reached a point at which a fundamental policy reform was required. In response, Mexico overhauled its monetary framework, shifting from fiscal dominance to monetary dominance and formally adopting an inflation-targeting regime in 2001 (Meza, 2019). While Mexico is a single-country case, its reform episode captures key features of successful monetary transitions in EMEs.

To evaluate the quantitative implications of the framework, we estimate the models using macro-international aggregates during Mexico's policy transition, employing a Bayesian approach. To identify and quantify the shocks driving inflation dynamics, it is not sufficient to rely on inflation alone. In an open-economy setting, exchange rate movements provide additional information that helps discipline the identification of aggregate demand forces, such as natural-rate shocks driven by productivity or foreign demand, when the output gap is unobserved. Accordingly, the observables are CPI inflation and nominal exchange rate depreciation vis-à-vis the U.S. dollar. We show below that the open-economy dimension is not merely a modeling convenience. When the model is re-estimated as a closed economy—using output growth in place of the exchange rate—the central bank's concern for robustness finds no support: the posterior of $\tilde{\theta}$ collapses toward zero, and the NRE specification becomes observationally equivalent to its RE counterpart.

The policy rate is not used as an observable. Instead, we assess external validity by comparing the model-implied optimal policy rate with the actual policy rate. Our analysis delivers the following main findings that highlight the quantitative relevance of a concern for robustness in post-crisis monetary policy. First, the estimated NRE model implies a substantial degree of con-

cern for robustness on the part of the central bank, suggesting that ambiguity about private-sector expectations plays a quantitatively important role in policy design. The implied detection error probability, a scale-free measure of the size of these doubts, is 0.071 at our sample length, close to the ten percent level conventionally targeted when the degree of concern for robustness is calibrated. Second, the NRE model significantly outperforms the RE model in terms of in-sample fit for the observables, as measured by the log marginal data density. Third, the monetary policy rate implied by the NRE model closely tracks Mexico’s observed policy rate, providing strong external validation of the framework. In contrast, the policy rate implied by the RE model fails to match the observed policy behavior. Importantly, the RE model does not satisfy external validity even when augmented with backward-looking price-setting behavior. This result demonstrates that conventional approaches to generating inflation persistence—through mechanical indexation to lagged inflation—are not sufficient to reconcile RE models with observed monetary policy behavior in the post-crisis period.

The strong concern for robustness in the NRE model gives rise to policy transmission mechanisms that differ fundamentally from those in the RE benchmark, in line with the channels emphasized in the ambiguity-aversion literature (e.g., [Woodford, 2010](#); [Kwon and Miao, 2017](#)). In response to a cost-push shock, domestic inflation in the NRE model rises by only about two-thirds as much as in the RE model—the imprint of a markedly more hawkish stabilization stance—while the exchange rate responds roughly twice as strongly, implying greater tolerance for exchange rate adjustments. These asymmetric policy responses reflect the central bank’s attempt to guard against worst-case distortions in inflation expectations.

The differences in policy transmission mechanisms across the two models are closely linked to the estimated structural parameters. In particular, the estimated NRE model features a substantially larger standard deviation of the cost-push shock relative to the RE model, whereas the productivity shock exhibits greater variability under RE. This reallocation of shock importance reflects the central bank’s concern for robustness in the presence of uncertainty about expectation formation and accompanies the asymmetric policy responses emphasized above.

Moreover, concern for robustness generates greater history dependence in inflation dynamics under NRE. Even in the absence of a backward-looking inflation term in the aggregate supply relation, the NRE model produces endogenous inflation inertia, whereas the RE model requires nearly half of all firms to exhibit backward-looking behavior to fit the data. This result directly addresses the criticisms of [Galí et al. \(2005\)](#) and [Rudd and Whelan \(2006\)](#) concerning the quantitatively modest role of backward-looking behavior and the ad hoc nature of inflation indexation: in the NRE framework, inflation persistence arises endogenously from the central bank’s concern for robustness, and the estimated results remain virtually unchanged when inflation indexation is eliminated entirely.

Taken together, these results indicate that the central bank’s concern for robustness—operating on the belief distortions that private agents actually hold—is what enables the NRE model to

generate a monetary policy rate closely aligned with the observed one. In the context of Mexico’s post-crisis monetary reforms, policymakers’ uncertainty about private-sector expectation formation plays a central role in shaping monetary policy behavior and inflation dynamics.

A series of additional exercises sharpens this interpretation. First, the identification of the concern for robustness is an open-economy phenomenon. As previewed above, the closed-economy re-estimation drives the posterior of $\tilde{\theta}$ toward zero; it also reverses the marginal data density ranking between the NRE and RE models, and the closed-economy model can reproduce the observed inflation persistence only by making nearly all firms backward-looking ($\delta = 0.96$). Openness and robustness are complements—openness without robustness fails to reproduce observed policy, and robustness without openness finds no support in the data. Second, the model’s empirical success cannot be attributed to the robust policy rule alone: an economy in which the central bank follows the same robustly optimal rule but private agents hold rational expectations largely matches the in-sample dynamics, yet fails to track the Bank of Mexico’s observed policy rate. The results thus favor the belief channel itself—the alignment of the central bank’s worst-case fears with the expectations that private agents actually hold. Finally, splitting the sample at Mexico’s formal adoption of inflation targeting in 2001 shows that both the estimated cost-push volatility and the empirical bite of the concern for robustness are concentrated in the early transition years: in the post-adoption period the data no longer meaningfully separate the two models, consistent with the gradual anchoring of expectations under the maturing regime. A complementary cross-country comparison points in the same direction: applied to Canada—an advanced economy with long-anchored expectations—the estimation identifies only a small concern for robustness ($\tilde{\theta} = 49.5$ against 1807), indicating that it does not manufacture robustness where expectations are anchored.

Related Literature. This paper is related to a growing body of literature studying the macroeconomic consequences of bounded rationality and deviations from full-information RE, particularly in the context of monetary policy design. Our primary contribution lies within the literature on robustly optimal monetary policy under ambiguity. To the best of our knowledge, this paper is the first to study robustly optimal monetary policy in an open-economy environment in which the policymaker faces ambiguity about private-sector NRE—corresponding to type-3 ambiguity in the classification of [Hansen and Sargent \(2012\)](#)—and to take this framework to the data. In particular, we are the first to estimate such a model using Bayesian methods, to discipline the framework quantitatively, and to apply it in an emerging-economy context to study policy predictions and conduct external validity tests of the model. Beyond characterizing robustly optimal policy in this environment, we show that the open-economy dimension is what renders the concern for robustness empirically identifiable in the first place—a result that, to our knowledge, is new to this literature. The contribution of the open-economy dimension is thus empirical rather than theoretical: openness does not add a channel through which the worst-case distortion operates—a result we establish formally in [Section 3](#)—but it supplies the exchange-rate margin along which that distortion leaves

an identifiable imprint on the data.

While [Woodford \(2010\)](#) characterizes robust monetary policy under type-3 ambiguity, the analysis is confined to a closed-economy setting. [Adam and Woodford \(2012\)](#) extend the analysis of robustly optimal policy under near-rational private-sector expectations to a fully microfounded New Keynesian model, and [Adam and Woodford \(2021\)](#) further develop this approach in an environment with a housing sector; both analyses likewise remain within a closed economy. Relatedly, [Benigno and Paciello \(2014\)](#) study optimal monetary policy when the policymaker faces doubts about the model, and [Karantounias \(2013, 2023\)](#) characterize optimal fiscal policy when either the private sector or the government distrusts the probability model, emphasizing—as we also find in our setting—the additional history dependence that a concern for robustness imparts to optimal policy. A large literature studies robust monetary policy in closed-economy New Keynesian models (see, for example, [Walsh, 2004](#); [Leitemo and Söderström, 2008b](#); [Dennis, 2010](#); [Levine and Pearlman, 2010](#); [Gerke and Hammermann, 2016](#)), and several papers extend robust policy analysis to open-economy environments (e.g., [Dennis et al., 2009](#); [Leitemo and Söderström, 2008a](#)). However, in these open-economy contributions, robustness typically reflects the policymaker’s doubts about the specification of exogenous shock processes or the underlying model, rather than uncertainty about private agents’ expectations. As a result, little is known about the features of robustly optimal monetary policy in open economies when policymakers are confident about the economic structure but uncertain about private-sector expectation formation. This paper fills this gap.

Our paper also contributes to the literature on monetary policy reforms in EMEs. As noted above, recent empirical work shows that countries with histories of high inflation continue to respond aggressively to inflation-expectations gaps even after adopting inflation-targeting regimes, highlighting the persistent influence of inherited expectations in shaping policy behavior ([Jácome et al., 2025](#)). We provide a structural interpretation of this evidence by embedding policymakers’ concerns about private-sector expectations into a monetary policy framework and quantifying their implications during a post-crisis policy transition.

More broadly, this paper relates to a large empirical and theoretical literature documenting systematic departures from full-information RE in expectation formation. Survey and experimental evidence show sizable deviations from RE in how private agents form forward-looking expectations, while a growing theoretical literature studies the macroeconomic and policy implications of such departures (e.g., [Coibion and Gorodnichenko, 2012](#); [Bordalo et al., 2020](#); [Afrouzi et al., 2023](#); [Angeletos and Lian, 2018](#); [Gabaix, 2020](#)). Our quantitative analysis also connects to studies that estimate New Keynesian models incorporating belief distortions, information frictions, ambiguity aversion, limited foresight, or diagnostic expectations on the part of private agents (e.g., [Ilut and Schneider, 2014](#); [Ilut and Saijo, 2021](#); [Bhandari et al., 2025](#); [Gust et al., 2020](#); [Bianchi et al., 2024](#); [L’Huillier et al., 2024](#); [Na and Yoo, 2025](#)). Unlike these studies, we focus on ambiguity faced by the policymaker rather than by private agents and estimate the degree of deviation from RE in an environment in which monetary policy is designed to be robust to private-sector NRE.

Finally, this paper relates to the literature on international business cycles and open-economy monetary policy. A central theme in this literature concerns the role of international financial market frictions, and a large body of work has studied their implications for optimal policy under RE. In contrast, our analysis deliberately abstracts from such frictions. This modeling choice is not intended to downplay their relevance, but rather to isolate the role of belief distortions and policymakers' concern for robustness within a parsimonious and transparent framework. By embedding ambiguity aversion into the canonical small open economy New Keynesian model of [Gali and Monacelli \(2005\)](#), we ensure that the mechanisms emphasized in the paper are not driven by auxiliary frictions but instead arise directly from uncertainty about private-sector expectation formation. This minimal structure facilitates identification of the policy implications of concern for robustness and allows for a clean comparison with the RE benchmark. Importantly, our approach follows earlier quantitative studies of Mexico that also adopt a parsimonious open-economy framework to focus on specific mechanisms of interest (e.g., [Aguiar and Gopinath, 2007](#); [Boz et al., 2011](#)).

Moreover, our empirical analysis focuses on the period from 1998 to 2007, spanning the post-Peso Crisis years up to the eve of the Global Financial Crisis, a relatively tranquil period for the Mexican economy. During this interval, financial frictions were less likely to be the primary drivers of business cycle fluctuations, making it a suitable laboratory for studying how monetary policy design responds to uncertainty about expectation formation. While incorporating richer financial structures remains an important avenue for future research, our results demonstrate that belief distortions alone can account for salient features of post-crisis monetary policy behavior.

The remainder of the paper is organized as follows. Section [2](#) introduces the belief-distortion framework underlying near-rational expectations, illustrates the resulting robustly optimal policy in a simple New Keynesian setting, and then develops the small open economy model and characterizes the corresponding robustly optimal policy. Section [3](#) presents an empirical analysis, estimating both the NRE and RE models for Mexico, examining the results and underlying mechanisms, and demonstrating the identifying role of the open-economy dimension. Section [4](#) collects complementary analyses and discussions. Section [5](#) concludes.

2 The Model

In this section, we first introduce the framework of belief distortions, then illustrate its implications for robustly optimal monetary policy in a simple closed-economy New Keynesian model, and finally develop the small open economy New Keynesian (SOE-NK) model on which our quantitative analysis is based.

2.1 Belief Distortion

The uncertainty in the economy is generated by the set of vectors of exogenous stochastic disturbances $\{\varepsilon_t\}_{t=0}^\infty$, where ε_t collects the model's i.i.d. standard normal innovations under the benchmark measure introduced below; its composition, for the small open economy, is specified in Section 2.3. We define $\varepsilon^t \equiv \{\varepsilon_0, \varepsilon_1, \dots, \varepsilon_t\}$ as the history of these periodic stochastic disturbances. We model the probability distortion by following Hansen and Sargent (2008). Let $(\Omega, \mathcal{F}, \mathcal{P})$ be a probability triple: Ω represents the sample space, $\mathcal{F}$ denotes its σ -field, and $\mathcal{P}$ is the associated probability measure. The benchmark, rational expectations (RE) operator, $\mathbb{E}[\cdot]$, is induced by measure $\mathcal{P}$, which is the rational measure for probabilities of exogenous states. However, we consider the possibility that expectations may not align with this rational measure, as they may be influenced by a potentially *different* measure $\hat{\mathcal{P}}$.¹ Let $p(\varepsilon)$ be the unconditional probability density of random vector ε (which has the same dimension as the entries of ε_t) under measure $\mathcal{P}$. Let $p(\varepsilon|\varepsilon^t)$ and $\hat{p}(\varepsilon|\varepsilon^t)$ denote the one-step-ahead probability densities for ε_{t+1} , induced by measures $\mathcal{P}$ and $\hat{\mathcal{P}}$, respectively, conditioned on date- t information. The likelihood ratio between the two conditional densities is

$$m_{t+1} = \frac{\hat{p}(\varepsilon|\varepsilon^t)}{p(\varepsilon|\varepsilon^t)} = \frac{\hat{p}(\varepsilon|\varepsilon^t)}{p(\varepsilon)},$$

where the second equality follows because the disturbances ε_{t+1} are i.i.d. under the benchmark measure $\mathcal{P}$, so that $p(\varepsilon|\varepsilon^t) = p(\varepsilon)$.² The likelihood ratio m_{t+1} is nonnegative and

$$\mathbb{E}[m_{t+1}|\varepsilon^t] = 1. \quad (2.1)$$

The one-step-ahead expectation of a random variable X_{t+1} induced by measure $\hat{\mathcal{P}}$ is then expressed as follows:³

$$\hat{\mathbb{E}}[X_{t+1}|\varepsilon^t] = \mathbb{E}[m_{t+1}X_{t+1}|\varepsilon^t].$$

Henceforth, we simply express an expectation based on date- t information as the expectation with subscript t , i.e., $\hat{\mathbb{E}}_t X_{t+1} \equiv \hat{\mathbb{E}}[X_{t+1}|\varepsilon^t]$ and $\mathbb{E}_t X_{t+1} \equiv \mathbb{E}[X_{t+1}|\varepsilon^t]$. The operator $\hat{\mathbb{E}}[\cdot]$ describes the potentially distorted forward-looking expectations in the economy. We use the relative entropy (Kullback-Leibler divergence) to quantify the difference between two probability measures, $\hat{\mathcal{P}}$ and $\mathcal{P}$. The divergence between one-period-ahead distorted belief and the rational belief is represented

¹We stipulate that $\hat{\mathcal{P}}$ must be absolutely continuous to $\mathcal{P}$.

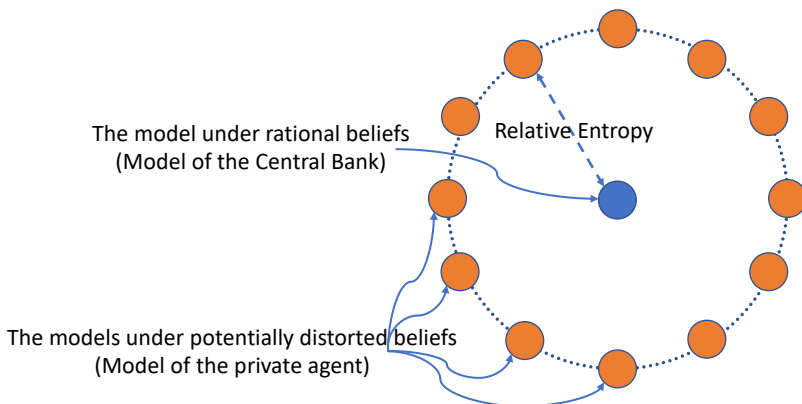
²The same unconditional representation is used in Kwon and Miao (2019).

³Following Hansen and Sargent (2008), we set $\mathcal{M}_0 = 1$ and recursively construct $\{\mathcal{M}_t\}$ such that $\mathcal{M}_{t+1} = m_{t+1}\mathcal{M}_t$, which implies $\mathcal{M}_{t+j} = \prod_{k=1}^j m_{t+k}$, and it is a martingale process that satisfies $\mathbb{E}[\mathcal{M}_{t+j}|\varepsilon^t] = \mathcal{M}_t$. Private agents' expectation $\hat{\mathbb{E}}[\cdot]$ of a random variable X_{t+j} induced by measure $\hat{\mathcal{P}}$ conditioned on date- t information can be expressed as the expectation induced by the measure $\mathcal{P}$ for the augmented random variable X_{t+j} :

$$\hat{\mathbb{E}}[X_{t+j}|\varepsilon^t] = \mathbb{E}\left[\frac{\mathcal{M}_{t+j}}{\mathcal{M}_t} X_{t+j} \middle| \varepsilon^t\right],$$

where $\frac{\mathcal{M}_{t+j}}{\mathcal{M}_t}$ represents the Radon-Nikodym derivatives, which completely summarize the belief distortions.

Figure 2: Entropy Diagram



by the relative entropy as follows:

$$\mathcal{R}_t \equiv \mathbb{E}_t m_{t+1} \ln m_{t+1},$$

which is always nonnegative based on Gibb's inequality. Given that policy outcomes depend on private-sector expectations, the central bank evaluates policy under uncertainty about the extent and persistence of these belief distortions. Specifically, the central bank seeks to minimize private agents' welfare loss but—unlike in the standard RE optimal policy problem—is concerned that the expectations it seeks to anchor may be distorted. Figure 2 illustrates this informational structure: the blue sphere denotes the central bank's RE benchmark, while the surrounding orange spheres represent private agents' belief models, with their distance from the benchmark measuring relative entropy. Because private agents act on their beliefs without internalizing potential model misspecification, the central bank faces uncertainty about how those beliefs are formed and therefore designs policy with a concern for robustness (ambiguity aversion).⁴ Following the robust control framework, we model this concern as a game against a hypothetical malevolent nature: nature chooses the belief distortion $\{m_{t+1}\}$ to maximize the central bank's loss, and the central bank chooses policy to minimize the loss under these worst-case beliefs.

2.2 Robustly Optimal Policy in a Simple New Keynesian Model

Before developing the full open-economy model, it is useful to illustrate the mechanism through which this concern shapes optimal policy in the simplest possible setting: the canonical closed-economy New Keynesian model with cost-push shocks and a dual mandate.⁵ The example previews the two features that drive our quantitative results—a more aggressive stance toward inflation stabilization and endogenous history dependence in policy—and motivates the conditionally linear

⁴This informational structure corresponds to *Type-3* ambiguity in Hansen and Sargent (2012).

⁵We thank an anonymous referee for suggesting this illustration, on which the presentation in this subsection builds.

solution method employed below. Suppose the central bank does not doubt its probability model of the exogenous shocks, but entertains the possibility that private-sector beliefs are described by a distorted measure, summarized by the one-step-ahead likelihood ratio m_{t+1} of Section 2.1. Following [Woodford \(2010\)](#), the central bank evaluates a *paternalistic* objective—the welfare loss under rational beliefs—augmented by an entropy penalty on the belief distortion. The robust policy problem is

$$\min_{\{\pi_t, x_t\}} \max_{\{m_{t+1}\}} \underbrace{\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{1}{2} \left(\pi_t^2 + \lambda_x (x_t - \bar{x})^2 \right)}_{\text{the welfare loss under RE}} - \underbrace{\theta \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t m_{t+1} \ln m_{t+1}}_{\text{the concern for robustness}}, \quad (2.2)$$

subject to the aggregate supply in which private-sector expectations are evaluated under the (potentially) distorted measure,

$$\pi_t = \kappa x_t + \beta \mathbb{E}_t m_{t+1} \pi_{t+1} + u_t, \quad (2.3)$$

and the restriction $\mathbb{E}_t m_{t+1} = 1$. Here π_t denotes inflation, x_t the output gap, and u_t an exogenous cost-push shock. The parameter $\beta \in (0, 1)$ is the subjective discount factor, $\kappa > 0$ governs the slope of aggregate supply, $\lambda_x > 0$ governs the weight on deviations of the output gap from its target value $\bar{x}$, and $\theta \in (0, \infty)$ reflects the degree of the central bank's concern for robustness. While the objective is convex with respect to π_t and x_t , it is concave with respect to m_{t+1} . As $\theta \rightarrow \infty$, the optimal behavior for the malevolent nature is to choose $m_{t+1} = 1$, corresponding to beliefs under RE; when θ is finite, the optimal choice of m_{t+1} induces a departure from RE beliefs. Note that m_{t+1} does not enter the first term of (2.2): since the central bank does not doubt the shock distribution, average losses are evaluated under the benchmark measure. The belief distortion matters only *indirectly*, through the forward-looking constraint (2.3), whose tightness depends on what the private sector expects. Assigning the multiplier $\beta^t \mu_t$ to (2.3), written so that the distorted expectation enters the Lagrangian with a positive coefficient,⁶ the inner maximization delivers the worst-case likelihood ratio

$$m_{t+1}^* = \frac{\exp\left(\frac{\beta \mu_t}{\theta} \pi_{t+1}\right)}{\mathbb{E}_t \exp\left(\frac{\beta \mu_t}{\theta} \pi_{t+1}\right)}, \quad (2.4)$$

while the first-order conditions with respect to the output gap and inflation are

$$\lambda_x (x_t - \bar{x}) = -\kappa \mu_t, \quad (2.5)$$

$$\mu_t = \pi_t + m_t^* \mu_{t-1}, \quad (2.6)$$

with the initial promise $\mu_{-1} \equiv 0$.

⁶That is, the constraint is written as $\kappa x_t + \beta \mathbb{E}_t m_{t+1} \pi_{t+1} + u_t - \pi_t = 0$.

Equations (2.4)–(2.6) contain the central intuition, organized around the multiplier μ_t , which plays two distinct roles. Through (2.5), it is the shadow price of the aggregate supply constraint, pinning down the output gap. Through (2.4), it is also what renders the belief distortion operative: the exponent in the worst-case likelihood ratio is proportional to $\beta\mu_t/\theta$, so at $\mu_t = 0$ we have $m_{t+1}^* = 1$ exactly and beliefs are rational, while the departure from RE grows with $|\beta\mu_t/\theta|$. The multiplier therefore measures how much leverage the central bank’s own position gives to the worst case.

The law of motion (2.6) describes how that position evolves. Under RE ($m_t^* = 1$), the multiplier simply cumulates realized inflation, $\mu_t = \pi_t + \mu_{t-1}$, so that $\pi_t = \mu_t - \mu_{t-1}$ —the familiar commitment dynamics, in which the multiplier is a running record of past inflation. Under robustness, the inherited promise μ_{t-1} is instead scaled by the worst-case ratio m_t^* , which varies with realized outcomes. Because m_t^* is increasing in π_t whenever $\mu_{t-1} > 0$, an adverse inflation realization scales the inherited promise *upward*, where under RE it would carry forward with weight exactly one. Past commitments are thus carried forward with state-dependent weight, providing an additional, endogenous source of history dependence in policy—and hence of inflation persistence—that operates without any backward-looking price setting. This state-dependent carry-forward of inherited promises echoes the central finding of Karantounias (2013, 2023), who shows—in settings where the ambiguity concerns the distribution of exogenous shocks rather than private-sector expectations—that the scaling of inherited multipliers by the equilibrium likelihood ratio is the source of the additional history dependence generated by a concern for robustness.

The direction of the distortion follows from (2.4). Consider $\mu_t > 0$, which by (2.5) is the case in which the output gap is held below target—a central bank disinflating the economy. The worst-case beliefs then place extra weight on *high* realizations of π_{t+1} . Through the aggregate supply relation (2.3), higher expected inflation raises current inflation for any given output gap, so achieving any given disinflation requires a larger output sacrifice than under RE. Worse, the effect feeds on itself: higher current inflation raises μ_t through (2.6), and a larger μ_t widens the distortion itself. The case $\mu_t < 0$ is the mirror image, with the worst case concentrating on *low*-inflation outcomes and the constraint binding against sustaining an expansion rather than against disinflating. In both cases the distortion tilts expectations in whichever direction raises the shadow cost of the forward-looking constraint.

Taken together, the two roles of the multiplier explain why a concern for robustness makes policy more aggressive. Because μ_t arms the worst case, it carries a cost beyond the promise-keeping role it plays under RE, and the central bank has a motive to keep it small. Because μ_t cumulates realized inflation, the only way to keep it small is to hold inflation tightly. The robust central bank does not avoid the output cost of doing so—by (2.3) it bears a *larger* output sacrifice than under RE—but accepts that cost to stop the multiplier, and with it the belief distortion, from building. It behaves *as if* the inflation–output tradeoff were systematically less favorable than under RE, and it optimally leans harder against inflation deviations.

Finally, the example clarifies why solving the problem requires the conditionally linear approach. Even though the objective is quadratic and the shocks may be normally distributed, the worst-case ratio m_t^* enters the multiplier dynamics (2.6) multiplicatively, so the system admits no closed-form linear solution. Restricting attention to policies that are linear conditional on the realized shocks—as proposed by [Woodford \(2010\)](#)—keeps the worst-case distortion log-normal and restores tractability. All of these elements carry over to the open-economy model developed next, where the forward-looking constraints additionally include the aggregate demand and exchange-rate relations, so the worst-case distortion operates on a richer set of margins.

2.3 A Small Open Economy New Keynesian Environment

We now develop the full open-economy model. As formalized in Section 2.1, private agents’ forward-looking expectations, denoted $\hat{\mathbb{E}}_t[\cdot]$, may be distorted; aside from this feature, the environment closely follows the canonical SOE-NK framework (e.g., [Galí and Monacelli, 2005](#)). The economy is infinitesimally small and exports domestic goods while importing foreign goods. Domestic firms use labor as the sole input, set prices in domestic currency, and face Calvo–Yun–style price rigidity. We also allow for backward-looking behavior in price setting, following [Galí and Gertler \(1999\)](#). In each period, a fraction of firms sets prices in a backward-looking manner as a function of lagged inflation. International financial markets are complete. Throughout, we present the log-linearized equilibrium conditions directly. The aggregate demand and open-economy relations follow the derivations in [Galí and Monacelli \(2005\)](#), with the rational expectation operator replaced by $\hat{\mathbb{E}}_t[\cdot]$; the aggregate supply relation with backward-looking price setting is derived in Appendix A.1. The log-linearized model is summarized by a set of linear equations describing domestic aggregate demand and supply, the links between domestic and international prices and quantities, as well as exogenous shock processes. Aggregate demand of the economy can be represented by the following equation:

$$x_t = \hat{\mathbb{E}}_t x_{t+1} - \frac{1}{\tau} \left(i_t - \hat{\mathbb{E}}_t \pi_{H,t+1} - \bar{r} r_t \right), \quad (2.7)$$

where x_t is the domestic output gap defined as the difference between actual output and its natural level under flexible prices, and $\pi_{H,t}$ is the net inflation rate of domestically produced goods (domestic inflation). The variable i_t is the short-term nominal interest rate, which serves as the central bank’s policy instrument. The parameter τ governs the slope of aggregate demand and is defined as follows:

$$\tau \equiv \frac{\sigma}{1 + \alpha(\omega - 1)}, \quad (2.8)$$

where $\sigma > 0$ governs the elasticity of intertemporal substitution in the private agents’ utility, $\alpha \in [0, 1]$ governs the degree of home bias ($1 - \alpha$), and ω governs the effect of changes in the terms

of trade (defined below) on output. The variable $\bar{r}\bar{r}_t$ is the natural real interest rate,

$$\bar{r}\bar{r}_t = \rho + \Lambda_{r,a}a_t + \Lambda_{r,y^*}y_t^*, \quad (2.9)$$

where a_t denotes the domestic productivity shock and y_t^* is the world output shock. Parameter $\rho = \beta^{-1} - 1$ is the steady-state real rate, and $\Lambda_{r,a}$ and Λ_{r,y^*} are coefficients that are functions of the structural parameters.

Aggregate supply incorporates both lagged domestic inflation and forward-looking expectations as follows:

$$\pi_{H,t} = \tilde{\beta}\hat{\mathbb{E}}_t\pi_{H,t+1} + \eta\pi_{H,t-1} + \kappa x_t + u_t, \quad (2.10)$$

where the random component u_t represents an exogenous cost-push shock that affects the aggregate supply of the economy, and

$$\tilde{\beta} \equiv \frac{\zeta\beta}{[\zeta + \delta(1 - \zeta(1 - \beta))]}, \quad \eta \equiv \frac{\delta}{[\zeta + \delta(1 - \zeta(1 - \beta))]}, \quad \kappa \equiv \frac{\zeta(1 - \delta)\kappa_0}{[\zeta + \delta(1 - \zeta(1 - \beta))]}, \quad (2.11)$$

with $\beta \in (0, 1)$ denoting the subjective discount factor, $\zeta \in [0, 1]$ as the Calvo–Yun parameter for nominal price rigidity, $\delta \in [0, 1]$ as the fraction of backward-looking firms,⁷ and $\kappa_0 > 0$ governing the slope of the aggregate supply curve, defined as follows:

$$\kappa_0 \equiv \frac{(1 - \beta\zeta)(1 - \zeta)(\tau + \varphi)}{\zeta}, \quad (2.12)$$

where φ denotes the inverse of the Frisch elasticity of labor supply. If the fraction of backward-looking firms, δ , equals zero, then $\tilde{\beta} = \beta$, $\eta = 0$, and $\kappa = \kappa_0$, so that the aggregate supply becomes purely forward-looking.

The log of the effective terms of trade, s_t , is defined as the difference between the log of the imported goods price and the log of the price of domestic goods, expressed as $s_t = p_{F,t} - p_{H,t}$. Complete international asset markets and the associated risk-sharing condition imply that s_t follows:

$$s_t = \tau(x_t + \tilde{y}_t - y_t^*), \quad (2.13)$$

where $\tilde{y}_t$ denotes the natural level of domestic output,

$$\tilde{y}_t = \Lambda_{y,0} + \Lambda_{y,a}a_t + \Lambda_{y,y^*}y_t^*, \quad (2.14)$$

⁷Following Galí and Gertler (1999), each backward-looking firm that resets its price in period t sets it equal to the average reset price of the previous period, corrected by lagged domestic inflation; see Appendix A.1 for the exact specification and the derivation of (2.10). This differs from the indexation scheme of Christiano et al. (2005), in which the prices of all firms that do *not* reoptimize are mechanically indexed to past inflation.

and $\Lambda_{y,0}$, $\Lambda_{y,a}$ and Λ_{y,y^*} are coefficients that depend on the structural parameters.⁸

Under the law of one price, the log effective exchange rate e_t satisfies $e_t = p_{F,t} - p_t^*$ up to a first-order approximation, with p_t^* being the log of the world effective price index. Using the definition of the terms of trade, nominal exchange rate depreciation, Δe_t —defined as the percentage change in the nominal exchange rate between periods t and $t - 1$ —can be expressed as:

$$\Delta e_t = \pi_{H,t} + \Delta s_t - \pi_t^*, \quad (2.15)$$

where Δ denotes the first-difference operator and π_t^* is the exogenous world inflation rate. A positive (negative) value of Δe_t implies a nominal depreciation (appreciation) of the domestic currency relative to the previous period.

The three shocks described above are assumed to follow first-order Markov processes:

$$u_{t+1} = \rho_u u_t + \sigma_u \varepsilon_{t+1}^u, \quad (2.16)$$

$$a_{t+1} = \rho_a a_t + \sigma_a \varepsilon_{t+1}^a, \quad (2.17)$$

$$y_{t+1}^* = \rho_{y^*} y_t^* + \sigma_{y^*} \varepsilon_{t+1}^{y^*}, \quad (2.18)$$

where the parameter ρ_k governs persistence, σ_k governs the standard deviation of each stochastic process, and ε_t^k is an i.i.d. standard normal innovation, for $k \in \{u, a, y^*\}$. The innovation vector of Section 2.1 thus specializes to $\varepsilon_t = (\varepsilon_t^u, \varepsilon_t^a, \varepsilon_t^{y^*})$.

Having described the open-economy environment, we next turn to the central bank's robustly optimal policy problem.

2.4 The Robustly Optimal Monetary Policy in the Open Economy

We now embed the concern for robustness of Section 2.1 into the full open-economy environment of Section 2.3. The central bank sets the nominal interest rate i_t to minimize the same paternalistic objective as in Section 2.2, with domestic inflation $\pi_{H,t}$ in place of π_t , now subject to the complete set of open-economy equilibrium relations. We define the central bank's robustly optimal monetary policy problem as the following game between the central bank and the malevolent nature:

Definition 2.1. *The robustly optimal monetary policy of the central bank solves the following mini-max optimization problem:*

$$\min_{\{\pi_{H,t}, x_t, i_t\}} \max_{\{m_{t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{1}{2} \left(\pi_{H,t}^2 + \lambda_x (x_t - \bar{x})^2 \right) - \theta \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t m_{t+1} \ln m_{t+1}, \quad (2.19)$$

subject to equations (2.7), (2.9), (2.10), (2.16), (2.17), (2.18), and (2.1).

Before turning to the solution, it is convenient to reparameterize the degree of the concern for robustness. Because a higher θ penalizes deviations from RE beliefs more heavily, a higher

⁸See Table E.8 in Appendix E for a detailed description of $\Lambda_{r,a}$, Λ_{r,y^*} , $\Lambda_{y,0}$, $\Lambda_{y,a}$, and Λ_{y,y^*} .

$\tilde{\theta} \equiv \theta^{-1} \in [0, \infty)$ can be interpreted as reflecting a greater degree of concern for robustness.⁹ In short, under the RE scenario, the central bank has no concern for robustness ($\tilde{\theta} = 0$), whereas under potentially distorted private-sector beliefs, the central bank has a concern for robustness ($\tilde{\theta} > 0$).

Conditionally Linear Commitment. As a benevolent planner, the central bank commits to its past policy promises, implying a history-dependent policy. The robustly optimal monetary policy problem defined in Definition 2.1 involves choosing a sequence $\{\pi_{H,t}, x_t\}_{t=0}^{\infty}$ that generally results in nonlinearity in the system. As illustrated in the simple example of Section 2.2, the source of this nonlinearity is the worst-case likelihood ratio itself, which multiplies the inherited policy promises in the multiplier dynamics (cf. equation (2.6)). We therefore adopt the conditionally linear commitment policy under a timeless perspective proposed by Woodford (2010), which restricts the policy rule to be linear in the current shock realization, conditional on the inherited promises. The worst-case distortion then remains log-normal and the equilibrium can be computed exactly. We begin by replacing the operator $\mathbb{E}_0$ in (2.19) with $\mathbb{E}_{-1}$, corresponding to expectations formed before the realization of the state in period 0. Rather than assuming that the policymaker selects a sequence $\{\pi_{H,t}, x_t\}$ for all $t \geq 0$, we focus on the choice of the sequence for $t \geq 1$, taking the initial commitment $\{\pi_{H,0}, x_0\}$ as given. We assume that the initial commitment takes a linear form:

$$\begin{bmatrix} \pi_{H,0} \\ x_0 \end{bmatrix} = \Phi_{-1} + \Gamma_{-1}\epsilon_0,$$

where $\Phi_{-1} \equiv [\Phi_{\pi_H, -1}, \Phi_{x, -1}]'$ and $\Gamma_{-1} \equiv \begin{bmatrix} \Gamma_{\pi_H, u, -1} & \Gamma_{\pi_H, a, -1} & \Gamma_{\pi_H, y^*, -1} \\ \Gamma_{x, u, -1} & \Gamma_{x, a, -1} & \Gamma_{x, y^*, -1} \end{bmatrix}$ characterize the initial commitment. We then focus on the following conditionally linear rule:

$$\begin{bmatrix} \pi_{H, t+1} \\ x_{t+1} \end{bmatrix} = \Phi_t + \Gamma_t \epsilon_{t+1}, \quad t \geq 0, \quad (2.20)$$

where $\Phi_t \equiv [\Phi_{\pi_H, t}, \Phi_{x, t}]'$ is stochastic and $\Gamma_t \equiv \begin{bmatrix} \Gamma_{\pi_H, u, t} & \Gamma_{\pi_H, a, t} & \Gamma_{\pi_H, y^*, t} \\ \Gamma_{x, u, t} & \Gamma_{x, a, t} & \Gamma_{x, y^*, t} \end{bmatrix}$ is deterministic. For any initial commitment $\{\Phi_{-1}, \Gamma_{-1}\}$, the central bank chooses $\{\Phi_t, \Gamma_t\}_{t=0}^{\infty}$. The initial commitment is self-consistent if

$$\begin{aligned} \Phi_{-1} &\stackrel{d}{=} \Phi_t \sim \Phi, \\ \Gamma_{-1} &= \Gamma_t = \Gamma, \end{aligned}$$

⁹We have used the parameter θ up to this point, following the convention in the ambiguity aversion literature. From this point onward, we use the transformed parameter $\tilde{\theta}$ to associate a higher value of the parameter with a greater degree of concern for robustness.

for all $t \geq 0$. This implies that Φ_{-1} and $\{\Phi_t\}_{t=0}^\infty$ follow the same unconditional distribution, and Γ_{-1} and $\{\Gamma_t\}_{t=0}^\infty$ are the same deterministic matrices. Consequently, the initial commitment $\{\Phi_{-1}, \Gamma_{-1}\}$ is self-consistent, and the robustly optimal policy has a time-invariant linear form.

To compute the equilibrium under the robustly optimal policy, we employ the solution method proposed by [Kwon and Miao \(2019\)](#). The system of equations for the linearized equilibrium of our model can be rewritten in the following form:

$$\begin{bmatrix} \mathbf{X}_{t+1} \\ \hat{\mathbb{E}}_t \mathbf{Y}_{t+1} \end{bmatrix} = \mathbf{A} \begin{bmatrix} \mathbf{X}_t \\ \mathbf{Y}_t \end{bmatrix} + \mathbf{B}i_t + \mathbf{C}\varepsilon_{t+1},$$

where $\mathbf{X}_t \equiv [1, \pi_{H,t-1}, u_t, a_t, y_t^*]'$ and $\mathbf{Y}_t \equiv [\pi_{H,t}, x_t]'$.¹⁰ The mini-max problem in (2.19) can be rewritten as the problem of choosing $\{\Phi_t, \Gamma_t\}_{t \geq 0}$, $\{\mathbf{X}_t, m_t\}_{t \geq 1}$, and $\{i_t\}_{t \geq 0}$ for a given initial commitment $\{\Phi_{-1}, \Gamma_{-1}\}$. The Lagrangian of the central bank's problem is then given by:

$$\begin{aligned} \mathcal{L} = & \mathbb{E}_{-1} \sum_{t=0}^{\infty} \beta^t \left\{ \frac{1}{2} \left(\pi_{H,t}^2 + \lambda_x (x_t - \bar{x})^2 \right) - \tilde{\theta}^{-1} m_{t+1} \ln m_{t+1} + \phi_t (\mathbb{E}_t m_{t+1} - 1) \right. \\ & \left. + [\mu'_{X,t+1}, \mu'_{Y,t}] \left(\begin{bmatrix} \mathbf{X}_{t+1} \\ \mathbb{E}_t m_{t+1} \mathbf{Y}_{t+1} \end{bmatrix} - \mathbf{A} \begin{bmatrix} \mathbf{X}_t \\ \mathbf{Y}_t \end{bmatrix} - \mathbf{B}i_t - \mathbf{C}\varepsilon_{t+1} \right) \right\}, \end{aligned} \quad (2.21)$$

where $\beta^t \phi_t$ is the Lagrange multiplier associated with constraint (2.1) and $\beta^t [\mu'_{X,t+1}, \mu'_{Y,t}]$ are vectors of Lagrange multipliers associated with the system of equations (2.7), (2.9), (2.10), (2.16), (2.17), and (2.18).

Solving this problem proceeds in several steps. First, the hypothetical malevolent nature chooses m_{t+1} to maximize the loss, yielding the worst-case belief m_{t+1} . Next, the policymaker chooses $\{\mathbf{X}_t, \Phi_t, \Gamma_t, i_t\}$ after substituting the solution of m_{t+1} into the Lagrangian. Because we focus on self-consistent policy, we additionally impose $\Gamma_t = \Gamma$. To obtain the solution for Γ , we begin with an initial guess for Γ and solve the first-order conditions, except for the first-order condition for i_t . The solution is then used to update Γ from the first-order condition for i_t . This process is repeated until convergence in Γ is achieved. We solve the resulting system of linear difference equations using the method of [Klein \(2000\)](#). From the solution, we obtain the following state-space representation:

$$\begin{bmatrix} \varepsilon_{t+1} \\ \mathbf{X}_{t+1} \\ \Phi_t \end{bmatrix} = \mathbf{H} \begin{bmatrix} \varepsilon_t \\ \mathbf{X}_t \\ \Phi_{t-1} \end{bmatrix} + \begin{bmatrix} \mathbf{I} \\ \mathbf{C}_X \\ \mathbf{0} \end{bmatrix} \varepsilon_{t+1}, \quad (2.22)$$

¹⁰The matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$, along with their partitions, are given in Appendix A.2.

and

$$\begin{bmatrix} i_t \\ \boldsymbol{\mu}_{Y,t} \\ \boldsymbol{\mu}_{X,t} \\ \mathbb{E}_t \boldsymbol{\mu}_{X,t+1} \end{bmatrix} = \mathbf{G} \begin{bmatrix} \boldsymbol{\varepsilon}_t \\ \mathbf{X}_t \\ \boldsymbol{\Phi}_{t-1} \end{bmatrix}, \quad (2.23)$$

where $\mathbf{I}$ is an identity matrix, $\mathbf{C}_X$ is a partition of $\mathbf{C}$, $\mathbf{0}$ is a zero matrix, and the matrices $\mathbf{H}$ and $\mathbf{G}$ govern the evolution of state variables and the policy rules for non-predetermined variables, respectively.

Finally, the worst-case beliefs, one-period-ahead expectations of variables under the worst-case beliefs, and the relative entropy in the worst-case scenario are given by

$$m_{t+1} = \exp \left(-\frac{1}{2} \tilde{\theta}^2 \boldsymbol{\mu}'_{Y,t} \boldsymbol{\Gamma} \boldsymbol{\Gamma}' \boldsymbol{\mu}_{Y,t} + \tilde{\theta} \boldsymbol{\mu}'_{Y,t} \boldsymbol{\Gamma} \boldsymbol{\varepsilon}_{t+1} \right), \quad (2.24)$$

$$\mathbb{E}_t m_{t+1} \mathbf{Y}_{t+1} = \boldsymbol{\Phi}_t + \tilde{\theta} \boldsymbol{\Gamma} \boldsymbol{\Gamma}' \boldsymbol{\mu}_{Y,t}, \quad (2.25)$$

$$\mathbb{E}_t m_{t+1} \ln m_{t+1} = \frac{1}{2} \tilde{\theta}^2 \boldsymbol{\mu}'_{Y,t} \boldsymbol{\Gamma} \boldsymbol{\Gamma}' \boldsymbol{\mu}_{Y,t}. \quad (2.26)$$

Equations (2.24)–(2.26) are the open-economy counterparts of the worst-case distortion in the simple example of Section 2.2, and they show precisely how the concern for robustness alters the equilibrium. The worst-case likelihood ratio (2.24) tilts probability mass toward innovation realizations aligned with $\boldsymbol{\Gamma}' \boldsymbol{\mu}_{Y,t}$ —that is, toward realizations that raise the shadow cost of the forward-looking constraints, as summarized by the multipliers $\boldsymbol{\mu}_{Y,t}$ inherited from the central bank's promises—and the magnitude of the tilt is proportional to the concern for robustness $\tilde{\theta}$. Equation (2.25) displays the resulting expectations wedge: under the worst-case beliefs, private-sector forecasts of the forward-looking variables $\mathbf{Y}_{t+1}$ deviate from the central bank's own forecast $\boldsymbol{\Phi}_t$ by $\tilde{\theta} \boldsymbol{\Gamma} \boldsymbol{\Gamma}' \boldsymbol{\mu}_{Y,t}$, so the forward-looking constraints bind as if expectations were systematically less favorable to the policymaker—the open-economy analogue of the tighter Phillips-curve tradeoff in Section 2.2. Equation (2.26) measures the size of the implied belief distortion, which the entropy penalty in the objective disciplines. Note, finally, that because $\boldsymbol{\mu}_{Y,t}$ is a function of the state inherited from past commitments, the expectations wedge is itself history dependent: the distortions the central bank guards against depend on its own past—the open-economy counterpart of the state-dependent carry-forward of promises in equation (2.6).

3 Empirical Analysis

In this section, we bring the theoretical framework in Section 2 to the data by estimating the model using the Mexican inflation rate and the nominal exchange rate. We consider two specifications: an NRE model in which the central bank's concern for robustness, $\tilde{\theta}$, is estimated, and an RE model in which $\tilde{\theta} = 0$.

Our empirical analysis proceeds in two steps. First, following standard practice, we evaluate each model’s fit to key unconditional second moments of the observables and the log marginal data density.¹¹ Second, we conduct an external-validity exercise by comparing the model-implied policy rate with the actual policy rate, as the policy rate is not included among the observables used in estimation. Together, these exercises address (i) what the data reveal about the magnitude of the central bank’s concern for robustness, $\tilde{\theta}$; (ii) how the NRE and RE models differ in their empirical predictions; and (iii) how well each model performs when evaluated against both standard business-cycle moments and the path of observed policy rates.

Belief Structure and Identification. Because the analysis moves beyond RE, it is important to state explicitly which probability models are involved when the framework is taken to the data. The following conceptually distinct objects must be distinguished: (a) the approximating model that the central bank has in mind; (b) the worst-case model that the central bank attributes to the private sector; (c) the actual model that the private sector uses in forming expectations; (d) the *true* model that generates the data; and (e) the model that the econometrician uses. Our baseline estimation assumes, first, that the central bank’s approximating model, the data-generating process, and the econometrician’s model coincide, i.e., (a) = (d) = (e); and second, that private agents actually form expectations under the worst-case beliefs, i.e., (b) = (c). In other words, in the baseline estimation the central bank’s worst-case assessment is borne out in equilibrium: the equilibrium law of motion that generates the observables is the one induced by the worst-case belief distortion (2.24) together with the robustly optimal policy. The estimated $\tilde{\theta}$ is accordingly identified from the imprint that this distortion leaves on the joint dynamics of inflation and the exchange rate. The alternative belief structure, in which the central bank designs robustly optimal policy while private agents actually hold RE—so that the actual and perceived laws of motion differ and the central bank’s concerns are unfounded—is examined in Section 4.1, where we re-estimate the model under this assumption and compare its performance.¹²

Data. We use CPI inflation, π_t , and nominal exchange rate depreciation vis-à-vis the U.S. dollar, Δe_t , as observables. In the model, CPI inflation satisfies (to a first-order approximation)

$$\pi_t = \pi_{H,t} + \alpha \Delta s_t, \quad (3.1)$$

while nominal depreciation is given by (2.15). We set world inflation π_t^* to zero. The world output process y_t^* is treated as exogenous and calibrated using U.S. output over the sample period, yielding $(\rho_{y^*}, \sigma_{y^*})$. Given the observables $\{\pi_t, \Delta e_t\}$, equations (3.1) and (2.15) discipline the joint dynamics of domestic inflation $\pi_{H,t}$ and terms-of-trade changes Δs_t , while the remaining equilibrium conditions pin down the implied output gap and policy rate.

¹¹The log marginal data density is computed using the modified harmonic mean estimator of Geweke (1999).

¹²A natural extension would allow the private sector and the central bank to hold separate robustness parameters; the two specifications we estimate are the polar cases of such a design along the private-sector dimension.

To construct the observables, we use quarterly CPI price indices and nominal exchange rates (vis-à-vis the U.S. dollar). The time spans of the CPI inflation rate and the depreciation rate are 1Q 1998–4Q 2007.¹³

The Linear State Space System for Estimation. We reproduce the system that governs equilibrium solution of the model. We denote $\boldsymbol{\xi}_t \equiv [\boldsymbol{\varepsilon}'_t, \mathbf{X}'_t, \boldsymbol{\Phi}'_{t-1}]'$ and $\boldsymbol{\chi}_t \equiv [i_t, \boldsymbol{\mu}'_{Y,t}, \boldsymbol{\mu}'_{X,t}, \mathbb{E}_t \boldsymbol{\mu}'_{X,t+1}]'$. Then equations (2.22) and (2.23) are rewritten compactly as

$$\begin{aligned}\boldsymbol{\xi}_{t+1} &= \mathbf{H}\boldsymbol{\xi}_t + \mathbf{S}^\epsilon \boldsymbol{\varepsilon}_{t+1}, \\ \boldsymbol{\chi}_t &= \mathbf{G}\boldsymbol{\xi}_t,\end{aligned}$$

respectively, where $\mathbf{S}^\epsilon \equiv [\mathbf{I}, \mathbf{C}'_X, \mathbf{0}]'$. Because the measurement equations for π_t and Δe_t involve lagged model variables, we use $[\boldsymbol{\xi}'_t, \boldsymbol{\xi}'_{t-1}]'$ as the state variables in the measurement system for estimation. The resulting measurement equation, allowing for measurement errors, is:

$$\begin{bmatrix} \pi_t \\ \Delta e_t \end{bmatrix} = \tilde{\mathbf{G}} \begin{bmatrix} \boldsymbol{\xi}_t \\ \boldsymbol{\xi}_{t-1} \end{bmatrix} + \begin{bmatrix} \sigma_\pi^{me} & 0 \\ 0 & \sigma_{\Delta e}^{me} \end{bmatrix} \cdot \begin{bmatrix} \epsilon_t^{me,\pi} \\ \epsilon_t^{me,\Delta e} \end{bmatrix}, \quad (3.2)$$

where $[\epsilon_t^{me,\pi}, \epsilon_t^{me,\Delta e}]'$ are orthogonal standard normal random variables, and $[\sigma_\pi^{me}, \sigma_{\Delta e}^{me}]'$ are the standard deviations of the measurement errors of the observables π_t and Δe_t , respectively.

The corresponding transition equation for estimation is:

$$\begin{bmatrix} \boldsymbol{\xi}_{t+1} \\ \boldsymbol{\xi}_t \end{bmatrix} = \tilde{\mathbf{H}} \begin{bmatrix} \boldsymbol{\xi}_t \\ \boldsymbol{\xi}_{t-1} \end{bmatrix} + \boldsymbol{\nu}_{t+1}. \quad (3.3)$$

Equations (3.2) and (3.3) together define the linear state-space system used for estimation. The details of $\tilde{\mathbf{G}}$, $\tilde{\mathbf{H}}$, and $\boldsymbol{\nu}_{t+1}$ are presented in Appendix B.

Pre-Set Parameters. The parameter σ , representing the curvature of the household utility function, is set to 1, following standard values in the literature. The Calvo–Yun nominal price rigidity parameter for domestic producers, ζ , is set to 0.75, indicating an average price adjustment interval of one year, which is also standard in the literature. For the rest of the world’s output, we use the overlapping U.S. output series for Mexico, performing an AR(1) regression. This results in a persistence parameter, ρ_{y^*} , of 0.84, and a standard deviation, σ_{y^*} , of 0.0045. The discount factor, β , is set to 0.975 to match the 10% annual country interest rate for emerging market countries (Uribe and Schmitt-Grohé, 2017). The home-bias related parameter, α , is calibrated to 0.279 to match the average import-to-GDP ratio of 27.9% over the sample period. We list the pre-set parameters in Table E.9.

Estimated Parameters. We estimate the remaining eight structural parameters $\boldsymbol{\Theta} \equiv [\tilde{\theta}, \delta, \omega, \varphi, \rho_u,$

¹³We intentionally disregard the periods after the 2008 global financial crisis, because our model does not have a zero lower bound restriction. The CPI price indices are seasonally adjusted by using the program package X13-ARIMA-SEATS.

Table 1: Posterior Distributions

Parameters	NRE		RE		NRE ₀		RE ₀	
	Mean	[5%, 95%]	Mean	[5%, 95%]	Mean	[5%, 95%]	Mean	[5%, 95%]
$\tilde{\theta}$	1807	[896, 3190]	-	-	1594	[831, 2708]	-	-
δ	0.10	[0.023, 0.21]	0.49	[0.14, 0.78]	-	-	-	-
ω	8.27	[6.16, 11.3]	4.31	[1.17, 7.62]	8.11	[5.99, 11.1]	6.47	[4.46, 9.14]
φ	1.97	[0.48, 4.80]	2.95	[0.33, 7.94]	1.35	[0.31, 3.44]	0.66	[0.10, 1.68]
ρ_u	0.32	[0.09, 0.59]	0.60	[0.12, 0.92]	0.38	[0.15, 0.62]	0.86	[0.75, 0.95]
σ_u	0.030	[0.016, 0.057]	0.013	[0.008, 0.023]	0.029	[0.016, 0.056]	0.013	[0.008, 0.022]
ρ_a	0.88	[0.77, 0.96]	0.88	[0.78, 0.96]	0.88	[0.77, 0.96]	0.87	[0.73, 0.96]
σ_a	0.031	[0.018, 0.049]	0.044	[0.027, 0.066]	0.027	[0.015, 0.044]	0.043	[0.029, 0.064]
Log MDD	209.57		192.07		212.82		192.00	

Note. The labels ‘NRE’ and ‘RE’ refer to the baseline models in Section 2. The labels ‘NRE₀’ and ‘RE₀’ refer to the NRE and RE models, which are estimated by imposing no backward-looking firms, i.e., $\delta = 0$. The posterior mean, 5th percentile, and 95th percentile were computed over one million draws from the MCMC chain.

Table 2: Implied Parameters Shaping Aggregate Demand and Supply

	NRE	RE	NRE ₀	RE ₀
τ	0.33	0.57	0.34	0.40
η	0.11	0.38	0	0
$\tilde{\beta}$	0.86	0.60	0.975	0.975
κ	0.16	0.08	0.15	0.09

Note. For the NRE and RE models, parameter values are obtained from the posterior means of δ , ω , and φ , together with other remaining calibrated parameters. In contrast, NRE₀ and RE₀ models impose $\delta = 0$ by construction.

$\sigma_u, \rho_a, \sigma_a]'$ and standard deviations of the measurement errors $[\sigma_\pi^{me}, \sigma_{\Delta e}^{me}]'$ using the likelihood-based Bayesian method.

The prior for $\tilde{\theta}$ is uniform on $[0, 10^4]$, reflecting an agnostic view over a wide range; the posterior distribution is well away from the RE boundary. For parameter δ , which governs the fraction of backward-looking firms, we assume a Beta (2,3) distribution as the prior. For parameters ω and φ , we assume Gamma (2,2) distributions as the prior.¹⁴ The prior distribution of the persistence of each shock process is assumed to follow a Beta (2,3) distribution. The prior distribution for 100 times the standard deviation of each shock process is assumed to follow a Gamma (0.25,2) distribution. Finally, for the standard deviations of the measurement errors of the observables, we impose uniform distributions with the maximum support to be 25% of the standard deviation of each observable as the prior. We use the random-walk Metropolis-Hastings algorithm with a 25 percent acceptance rate to generate 1 million draws of MCMC chains, which we then use to assess the posterior distributions. We place the details of the prior distributions of the eight structural parameters in Table E.10.

¹⁴Rather than estimating the shape parameters τ , $\tilde{\beta}$, η and κ directly, we estimate δ , ω and φ , as they jointly affect the shape parameters through the parametric relationships (2.8), (2.11), and (2.12). In Table 2, we report the implied shape parameters from the posterior mean estimates.

Table 1 reports posterior distributions and log marginal data densities for the baseline NRE and RE models, as well as the corresponding versions that impose $\delta = 0$ (NRE₀ and RE₀). The NRE estimates imply a sizable degree of concern for robustness: the posterior mean of $\tilde{\theta}$ is 1807 with a 90% credible interval of [896, 3190], placing it far from the RE limit of zero. The two models also differ sharply in the extent of backward-looking price setting. Under NRE, the posterior mean of δ is 0.10, whereas the RE model has $\delta = 0.49$. Consistent with this contrast, imposing $\delta = 0$ leaves the NRE posterior largely unchanged (NRE₀ versus NRE), while it induces sizable shifts in key parameters under RE (RE₀ versus RE), indicating that the RE benchmark relies more heavily on the backward-looking channel.

Beyond $\tilde{\theta}$ and δ , several parameters differ meaningfully across the NRE and RE models. The open-economy parameter ω is substantially larger under NRE (8.27) than under RE (4.31), implying a stronger role for terms-of-trade movements in shaping domestic dynamics. The posterior mean of φ is 1.97 under NRE and 2.95 under RE. The estimated cost-push shock is less persistent but more volatile under NRE ($\rho_u = 0.32$, $\sigma_u = 0.030$) than under RE ($\rho_u = 0.60$, $\sigma_u = 0.013$).¹⁵ The persistence of the productivity shock is similar across models ($\rho_a = 0.88$ in both), while its volatility is smaller under NRE ($\sigma_a = 0.031$) than under RE ($\sigma_a = 0.044$).

Furthermore, in terms of in-sample fit, the NRE specification overwhelmingly outperforms the RE benchmark. The difference in log marginal data density is economically large (209.57 vs. 192.07), corresponding to decisive Bayesian evidence in favor of NRE. Importantly, this ordering is preserved—and even strengthened—when backward-looking price setting is eliminated (212.82 for NRE₀ vs. 192.00 for RE₀), indicating that the superior performance of NRE does not hinge on mechanical inflation persistence but instead reflects the central bank’s concern for robustness.

Table 2 reports the implied parameters governing the shapes of aggregate demand (τ) and aggregate supply (η , $\tilde{\beta}$, κ), constructed from the posterior means of δ , ω , and φ in Table 1 using the parametric relationships in (2.8), (2.11), and (2.12). Two main patterns emerge. First, the implied slopes differ markedly between the NRE and RE models. The NRE model features a more interest-sensitive aggregate demand relation, with a lower value of τ (0.33 versus 0.57), and a substantially more forward-looking aggregate supply relation. In particular, the weight on lagged inflation is much smaller under NRE ($\eta = 0.11$) than under RE ($\eta = 0.38$), while the forward-looking component is correspondingly larger ($\tilde{\beta} = 0.86$ versus 0.60). At the same time, the slope of aggregate supply with respect to the output gap is steeper under NRE ($\kappa = 0.16$) than under RE ($\kappa = 0.08$). Second, imposing $\delta = 0$ leaves the implied parameters under NRE largely unchanged, whereas it induces substantial shifts under RE—most notably in τ .

¹⁵Without the backward-looking firms ($\delta = 0$), RE₀ must substitute exogenous shock persistence (ρ_u goes from 0.60 to 0.86) to reproduce inflation persistence, whereas NRE₀ need not (ρ_u goes from 0.32 to 0.38), because it generates persistence endogenously.

Table 3: Second Moments: Data and Models

Moments	Data	NRE	RE	NRE ₀	RE ₀
$\sigma(\pi_t)$	1.09	0.81	1.23	0.83	1.22
$\sigma(\Delta e_t)$	3.12	3.30	3.10	3.28	2.94
$\rho(\pi_t, \Delta e_t)$	0.28	0.23	0.31	0.26	0.39
$\rho(\pi_t, \pi_{t-1})$	0.92	0.65	0.37	0.69	0.23
$\rho(\pi_t, \pi_{t-2})$	0.80	0.44	0.09	0.49	0.07
$\rho(\Delta e_t, \Delta e_{t-1})$	0.13	-0.09	0.05	-0.06	-0.02
$\rho(\Delta e_t, \Delta e_{t-2})$	0.07	-0.10	-0.03	-0.08	-0.03

Note. The second moments of the models are computed using the posterior means of the parameters for each model.

3.1 Predictions of the Estimated Models

We now discuss the predictions of the estimated models, focusing on their in-sample performance in matching the second moments of the observables and their out-of-sample performance (external validity), as measured by the model-implied monetary policy rates and their alignment with the actual policy rate.

Second Moments. Table 3 presents the empirical and theoretical second-order moments for CPI inflation and nominal exchange rate depreciation, comparing the observables with the predictions of the models. The results indicate that both the NRE and RE models perform similarly in terms of fitting the unconditional standard deviations and cross-correlations. However, the NRE model is more successful in predicting inflation persistence, as reflected in the higher autocorrelations of inflation ($\rho(\pi_t, \pi_{t-1})$ and $\rho(\pi_t, \pi_{t-2})$) relative to the RE model, even though aggregate supply in the NRE model is significantly more forward-looking, as indicated by the estimates of δ , η , and $\tilde{\beta}$ in Tables 1 and 2. While nominal exchange rate depreciation exhibits little persistence in the data and is reasonably captured by both models, the key quantitative distinction between the models in Table 3 arises from their implications for inflation dynamics.

The results from the NRE₀ and RE₀ models further reinforce this conclusion. The NRE₀ specification—despite completely shutting down the backward-looking component in the aggregate supply relation—continues to generate inflation persistence comparable to, and even slightly stronger than, that of the baseline NRE model. By contrast, the RE₀ model performs markedly worse than the RE benchmark in matching observed inflation persistence. Combined with the evidence from Tables 1 and 2, which show no meaningful differences in estimated parameters between the NRE and NRE₀ models, these results indicate that the NRE framework does not rely on backward-looking price indexation to fit the data.

This finding is important in light of long-standing critiques of lagged inflation terms in New Keynesian models, which—while often improving empirical fit—are frequently viewed as ad hoc

(Galí et al., 2005). In the NRE model, inflation persistence arises endogenously from the interaction between the central bank’s concern for robustness and private-sector expectations, rather than from mechanical indexation to past inflation. In contrast, the RE model depends heavily on backward-looking pricing behavior to reproduce inflation persistence. Thus, the NRE framework provides a quantitatively successful and conceptually disciplined alternative to backward-looking inflation mechanisms.

Prediction on Monetary Policy Rates. We now examine how closely the optimal monetary policy rates generated by the estimated models resemble the actual data.¹⁶ Using the estimated models and observables, we extract the historical paths of exogenous shocks through the Kalman smoother and feed them into the models to obtain the monetary policy rates implied endogenously by each model. Because the monetary policy rate is not used as an observable in estimation, this comparison provides an out-of-sample evaluation of model fit and serves as an external-validity check of the framework.

Figure 3 plots the actual monetary policy rate in Mexico together with the policy rate paths implied by the models. The NRE model clearly outperforms the RE benchmark in terms of predictive accuracy. Over the entire sample, the policy rate implied by the NRE model closely tracks the actual data, whereas the RE model exhibits substantial deviations. Quantitatively, the RMSE of the policy rate path is 0.0295 under NRE, compared with 0.0788 under RE, corresponding to a reduction of approximately 63%.

The contrast becomes even sharper when backward-looking price setting is shut down. The NRE_0 model performs almost identically to the baseline NRE model, whereas the RE_0 model deteriorates further relative to the RE benchmark. The RMSE of the policy rate path is 0.0301 under NRE_0 , compared with 0.0910 under RE_0 , implying a reduction of approximately 67%.

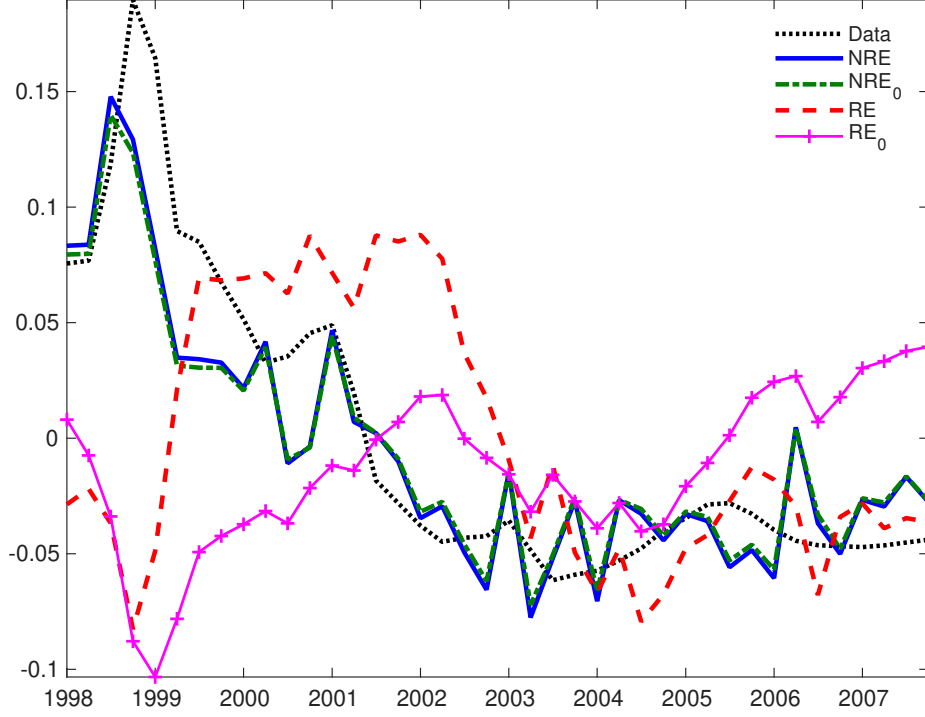
3.2 Inspecting Mechanism

In this section, we investigate the mechanisms underlying the models’ empirical predictions and clarify their economic interpretation. To begin, we note that even under a concern for robustness, optimal monetary policy retains a sharp focus on specific sources of distortion. In particular, robustness does not alter which shocks generate policy trade-offs; rather, it reshapes how the central bank responds to those trade-offs once they arise. To formalize this insight, we begin by establishing a lemma that characterizes the normative structure of the robustly optimal policy.

Lemma 3.1. *For any $\tilde{\theta} \in \mathbb{R}_+$, the robustly optimal monetary policy calls for zero domestic inflation and a zero output gap in response to domestic productivity shocks a_t and world output shocks y_t^* , i.e., $\Gamma_{\pi_H, a} = \Gamma_{x, a} = \Gamma_{\pi_H, y^*} = \Gamma_{x, y^*} = 0$.*

¹⁶We use the Mexican overnight interbank rate as a proxy for the actual monetary policy rate. All series are on a quarterly frequency and have been demeaned.

Figure 3: Monetary Policy Rates: Data and Model Predictions



Note. The black dotted line denotes the monetary policy rate in the data. The blue solid line (labeled NRE) and the red dashed line (labeled RE) show the policy rate paths predicted by the estimated NRE and RE models, respectively. The green dash-dotted line (labeled NRE_0) and the magenta line with crosses (labeled RE_0) show the predicted policy rate paths from the estimated NRE and RE models without backward-looking firms (i.e., $\delta = 0$), respectively. All series are demeaned and annualized. The RMSEs for NRE, RE, NRE_0 , and RE_0 are 0.0295, 0.0788, 0.0301, and 0.0910, respectively.

Table 4: Equilibrium Responses to the Cost-Push Shock Under the Optimal Policy

	NRE	RE	NRE_0	RE_0
$\Gamma_{\pi_H, u}$	0.0088	0.0143	0.0087	0.0141
$\Gamma_{x, u}$	-0.1152	-0.0493	-0.1138	-0.0541

Note. The equilibrium responses are computed using the posterior means of the parameters of each model.

Lemma 3.1 implies that, regardless of the degree of concern for robustness, only the cost-push shock u_t affects domestic inflation $\pi_{H,t}$ and the output gap x_t under the optimal policy. Domestic productivity shocks a_t and world output shocks y_t^* influence the natural real interest rate $\bar{r}_t$ (equation (2.9)), but their effects on aggregate demand are fully offset by policy, even in the presence of distorted private-sector expectations. As a result, there is no trade-off between stabilizing domestic inflation and the output gap—the *divine coincidence*. These shocks nevertheless affect the economy through the natural level of output $\tilde{y}_t$, thereby influencing terms-of-trade and nominal exchange rate dynamics (equations (2.13) and (2.15)).

Equilibrium Responses to the Cost-Push Shock. Guided by Lemma 3.1, we focus on the optimal policy responses to the cost-push shock. Table 4 reports $\Gamma_{\pi_H,u}$ and $\Gamma_{x,u}$, the equilibrium impact responses of domestic inflation and the output gap, respectively, to a one-standard-deviation cost-push innovation under the optimal policy, and reveals a stark contrast between the NRE and RE models. Under NRE, domestic inflation responds markedly less to the shock ($\Gamma_{\pi_H,u} = 0.0088$ against 0.0143, a reduction of roughly 40 percent), while the output gap absorbs substantially more of it ($\Gamma_{x,u} = -0.1152$ against -0.0493 , more than double). The contrast reflects a distinctly more hawkish stabilization stance under robustness: less inflation variability is purchased with more output-gap variability, a fundamental reallocation of stabilization effort. The mechanism is the one described in Section 2.2: containing inflation limits the growth of the policy multiplier, and hence of the worst-case belief distortion in equation (2.6).

Figure 4 illustrates the economic mechanism underlying these differences. In the RE model, the optimal inflation response $\Gamma_{\pi_H,u}$ increases linearly with the size of the cost-push shock σ_u , reflecting the principle of *certainty equivalence*. This principle breaks down under NRE. Relative to the RE counterfactual (RC) that shares the same parameter estimates but sets $\tilde{\theta} = 0$, the NRE policy exhibits a concave response: as σ_u increases, $\Gamma_{\pi_H,u}$ rises less than proportionally and eventually declines.¹⁷ This pattern indicates that a central bank concerned with robustness becomes increasingly reluctant to scale up its inflation response in the presence of large shocks.

Consequently, the NRE model reallocates stabilization toward inflation and away from the output gap. As documented in Table 1, this behavior is driven jointly by a high estimated concern for robustness ($\tilde{\theta}$) and a large estimated volatility of cost-push shocks (σ_u), leading the central bank to adopt a markedly more hawkish inflation stabilization stance than under RE—the quantitative counterpart of the mechanism illustrated in Section 2.2, whereby the worst-case distortion tilts private-sector expectations in precisely the direction that worsens the inflation–output tradeoff.

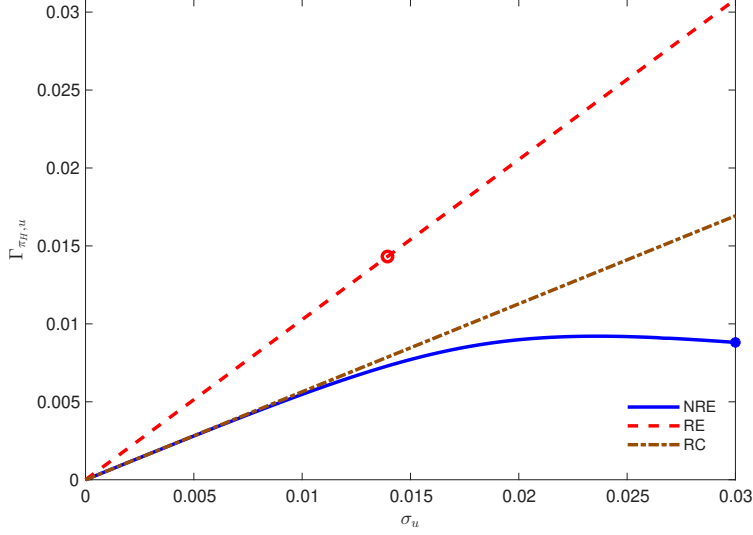
This reallocation has direct implications for exchange-rate dynamics. Because nominal depreciation responds to both domestic inflation and movements in the output gap (equations (2.13) and (2.15)), the initial response of Δe_t closely mirrors $\Gamma_{x,u}$. Quantitatively, the larger magnitude of $\Gamma_{x,u}$ relative to $\Gamma_{\pi_H,u}$, combined with the muted inflation response under NRE, implies that exchange-rate adjustment operates primarily through the output-gap channel.

Table 4 further demonstrates that the NRE₀ model delivers equilibrium responses nearly identical to those of the baseline NRE model. This stability underscores that the robustness channel, rather than backward-looking pricing, is the primary driver of optimal policy behavior in the NRE framework. By contrast, the RE₀ model exhibits a noticeably larger output-gap response than the RE benchmark, indicating that backward-looking components play a quantitatively important role in sustaining the policy stance under RE.

The following proposition shows how the degree of concern for robustness affects the optimal

¹⁷Figure E.10 in Appendix E plots the policy-rate path implied by the RC—the estimated NRE parameters with $\tilde{\theta} = 0$: it is close to that of the RE model, with a comparable failure to track the actual rate.

Figure 4: Equilibrium Inflation Response $\Gamma_{\pi_H,u}$: Graphical Illustration



Note. The blue solid and red dashed lines depict the equilibrium inflation response $\Gamma_{\pi_H,u}$ as a function of the standard deviation of the cost-push shock, σ_u , with all other parameters fixed at their posterior means from the estimated NRE and RE models, respectively. The circle on each line indicates the posterior mean of σ_u for the corresponding model. The brown dash-dot line depicts $\Gamma_{\pi_H,u}$ under the RE counterfactual (RC), which uses the estimated NRE model parameters but sets $\tilde{\theta} = 0$.

dynamics of the domestic inflation rate:

Proposition 3.2. *Consider $\delta = 0$. In equilibrium with the robustly optimal monetary policy, the dynamics of domestic inflation is determined by the following equation:*

$$\pi_{H,t} = \rho_{\pi_H} \pi_{H,t-1} + \Gamma_{\pi_H,u} \varepsilon_t^u - \rho_{\pi_H} (1 - \rho_u) \sigma_u \sum_{j=0}^{\infty} \rho_u^j \varepsilon_{t-j-1}^u, \quad (3.4)$$

where $\rho_{\pi_H} \in (0, 1)$ increases as $\tilde{\theta}$ increases.

Proposition 3.2 isolates the effect of robustness on endogenous inflation persistence by imposing $\delta = 0$. As the concern for robustness increases, the autoregressive coefficient ρ_{π_H} rises, strengthening the dependence of current inflation on its own past. Moreover, when the cost-push shock is persistent ($\rho_u > 0$), this increase in ρ_{π_H} amplifies the contribution of past cost-push shocks to current inflation through the moving-average component of (3.4). Taken together, these effects imply that heightened concern for robustness induces more history-dependent inflation dynamics relative to the RE benchmark. This insight echoes the conclusions of [Woodford \(2010\)](#), which are derived in a closed-economy setting. Proposition 3.2 extends this logic by highlighting how shock persistence amplifies history dependence under robust monetary policy.

These results imply that inflation inertia in the NRE model emerges *endogenously* from robust monetary policy, even in the absence of backward-looking price setting, as also reflected in the model's fit to inflation autocorrelations in Table 3. This mechanism is conceptually distinct from

Table 5: Decomposition of Unconditional Variances

Model Series/shock	NRE			RE			NRE ₀			RE ₀		
	Cost	TFP	World	Cost	TFP	World	Cost	TFP	World	Cost	TFP	World
CPI Inflation	75.4	21.7	0.45	62.1	36.1	0.38	77.3	19.7	0.52	58.0	40.2	0.43
Depreciation Rate	78.7	16.7	0.34	24.1	72.5	0.77	79.4	15.9	0.42	7.90	88.7	0.94

Note. The labels “Cost”, “TFP”, and “World” refer to the cost-push shock, domestic productivity shock, and world output shock, respectively. The variance decompositions are performed using the posterior means of parameters of the models. Shares are in percent. The sum of the contributions from each shock does not equal 100, and the remaining portions are due to measurement errors.

the traditional backward-looking channel: while the latter generates inertia mechanically through firms’ pricing rules, robustness gives rise to persistence through a breakdown of certainty equivalence in policy design—the feedback from inherited policy promises to worst-case beliefs formalized in equation (2.6) and, in the full model, in the history-dependent expectations wedge of equation (2.25). The small posterior estimates of δ in the NRE model further indicate that backward-looking behavior is not the dominant source of inflation persistence in this environment.

Shock Decompositions. The sharply different policy stances and transmission mechanisms implied by the NRE and RE models lead them to interpret the data differently when identifying the sources of macroeconomic fluctuations. To clarify the mechanisms underlying these contrasting policy implications, we examine shock decompositions. This analysis reveals systematic differences in how the two frameworks attribute fluctuations in observables to underlying shocks, with particularly pronounced contrasts in the role assigned to cost-push disturbances.

Table 5 reports the unconditional variance decompositions of the two observables. Across all models, world output shocks play only a negligible role in explaining fluctuations in either variable. Under the NRE model, cost-push shocks dominate fluctuations in both variables, accounting for 75.4% of inflation variance and 78.7% of exchange-rate variance. By contrast, the RE model assigns a much smaller role to cost-push shocks—especially for depreciation—while attributing the bulk of exchange-rate fluctuations to productivity shocks (72.5%). This contrast becomes even sharper when backward-looking pricing is shut down. The NRE₀ model delivers decompositions almost identical to the baseline NRE model, whereas the RE₀ model shifts further toward a TFP-driven interpretation, with productivity shocks explaining 88.7% of exchange-rate variance.

These differences in variance attribution have direct implications for the interpretation of monetary policy dynamics. To illustrate this link, we decompose the model-implied policy rates using the estimated shock histories. Figure 5 shows that the divergence between the NRE and RE models extends beyond overall fit to the underlying sources of policy-rate movements. In the NRE model, cost-push shocks emerge as the primary driver of the historical dynamics of the policy rate. In contrast, the RE model assigns a much more limited role to cost-push shocks and instead attributes policy movements largely to productivity shocks. This pattern persists when backward-looking

pricing is removed. The NRE_0 model closely mirrors the NRE benchmark, while the RE_0 model exhibits a stark reallocation toward productivity-driven policy dynamics. As a result, the RE_0 model fails to reproduce the observed path of the policy rate.

Viewed through this lens, these decompositions reveal that the key distinction between the two frameworks lies in how policy-relevant shocks are interpreted. Under NRE, the central bank’s concern for robustness leads policy dynamics to be driven primarily by cost-push disturbances, consistent with the observed behavior of the policy rate during Mexico’s post-crisis transition. By contrast, the RE framework interprets policy movements largely through productivity shocks, a mapping that fails to reproduce the data even when backward-looking pricing is introduced. From this perspective, the success of the NRE model reflects not an auxiliary role for backward-looking behavior, but a fundamentally different reading of the shocks that matter for policy, shaped by optimal policy design under a concern for robustness.

3.3 The Role of the Open-Economy Dimension

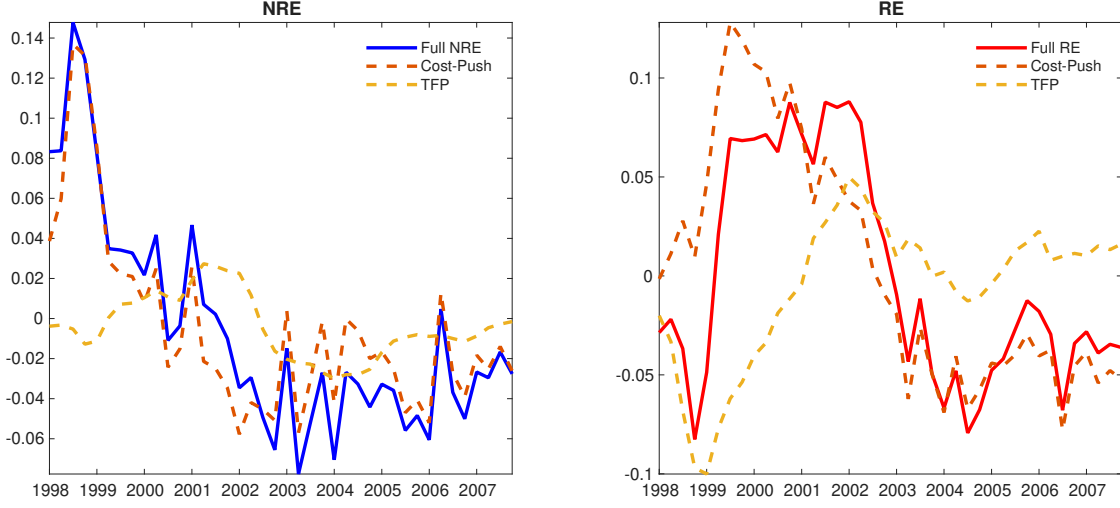
The mechanism through which a concern for robustness induces a more hawkish inflation-stabilization stance and endogenous history dependence is, qualitatively, not specific to the open economy: as Section 2.2 illustrates, it operates already in a closed-economy setting, in line with the results of Woodford (2010). A natural question, therefore, is what the open-economy dimension contributes to our analysis. In this subsection we show that the answer is empirical and twofold: the open-economy structure is what identifies the central bank’s concern for robustness in the data, and the concern for robustness and openness are complements—neither alone reproduces observed policy behavior.

To establish this point, we estimate a closed-economy variant of the model. When $\alpha = 0$ (complete home bias), our framework is isomorphic to a standard closed-economy New Keynesian model: CPI inflation coincides with domestic inflation, the terms of trade and the exchange rate drop out of the equilibrium system, and the world-output shock is absent. Since the depreciation rate is no longer defined, we replace it with output growth as the second observable, so that the closed-economy estimation uses $\{\pi_t, \Delta y_t\}$; the belief-distortion framework, the robustly optimal policy problem, the priors, and the Bayesian procedure are otherwise identical to the baseline estimation. Appendix C presents the closed-economy equilibrium conditions and the associated state-space system in detail. We emphasize that the exercise necessarily changes the model structure and the observable set jointly: the exchange-rate observable exists only in the open economy, and providing this additional observable margin is precisely part of what the open-economy structure contributes.

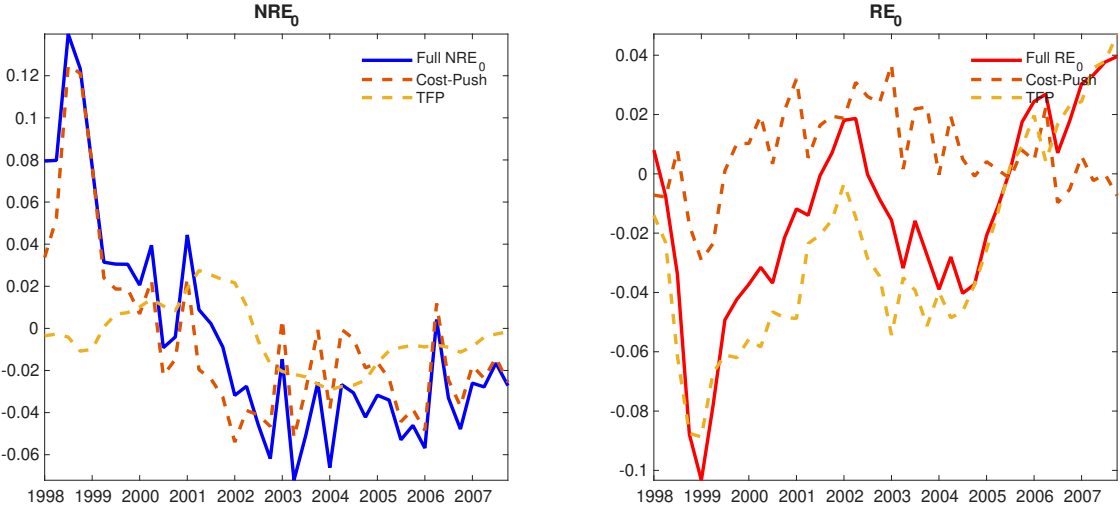
Table 6 summarizes the results; full posterior distributions are reported in Table E.11 in Appendix E. Three findings stand out. First, the estimated concern for robustness essentially vanishes. In the baseline closed-economy specification, the posterior mean of $\tilde{\theta}$ is 0.076, with a 90% credible interval of [0.004, 0.233], compared with 1807 and [896, 3190] in the open economy; with-

Figure 5: Monetary Policy Rates in the Models Conditional on Shocks

(a) Baseline



(b) No Backward-Looking ($\delta = 0$)



Note. The blue and red solid lines in panel (a) represent the monetary policy rates predicted by the NRE and RE models, respectively, while the blue and red solid lines in panel (b) represent the policy rates predicted by the NRE_0 and RE_0 models, respectively. These series are identical to those shown in Figure 3. The dashed lines labeled ‘Cost-Push’ and ‘TFP’ correspond to predictions conditional only on the cost-push shock and domestic productivity shock, respectively. The world output shock is quantitatively negligible.

out backward-looking price setting, the posterior mean is 34.33 with a 90% credible interval of [2.42, 98.30], compared with 1594 and [831, 2708]. The NRE and RE models deliver essentially identical parameter estimates. Indeed, the ranking by log marginal data density flips: whereas the open-economy data decisively favor the NRE model, the closed-economy data favor the RE model (253.21 for NRE versus 265.37 for RE). The flip is consistent with the standard Bayesian penalty for estimating a parameter, $\tilde{\theta}$, that the data determine to be economically irrelevant.

Second, the closed-economy model can mimic the observed inflation persistence only by all but

eliminating the forward-looking channel in aggregate supply. The posterior mean of δ rises to 0.96 (NRE) and 0.97 (RE)—nearly all firms must be backward-looking—in sharp contrast to $\delta = 0.10$ in the open-economy NRE model, where persistence arises endogenously from the robustly optimal policy. When this backward-looking channel is shut down ($\delta = 0$), the closed-economy model fails entirely: the policy-rate RMSE rises to 0.152, and, as Figure 6 shows, the implied policy rate moves in the wrong direction during the volatile opening years of the sample. Relatedly, the closed-economy model with $\delta = 0$ pushes the cost-push persistence toward a unit root ($\rho_u = 0.94$), substituting shock persistence for the missing indexation.

Third, even at its best, the closed-economy model falls short of the open-economy NRE benchmark in predicting the actual monetary policy rate. As Figure 6 shows, the baseline closed-economy model captures the broad downward path of the policy rate that accompanied the post-crisis disinflation, but substantially understates the amplitude, under-predicting the early-sample peak and implying near-zero rate deviations after 2001, when actual rates remained well below the sample mean; its RMSE (0.0405) exceeds that of the open-economy NRE model (0.0295) by more than one-third. Notably, openness alone does not deliver this improvement mechanically: the closed-economy baseline in fact outperforms the open-economy RE model (0.0788). It is the *combination* of the open-economy structure and the concern for robustness that reproduces observed policy behavior—openness without robustness fails, and robustness without openness finds no support in the data.

The economics behind the identification result is an asymmetry in what the observables reveal. As Section 3.2 documents, the concern for robustness reallocates the response to cost-push shocks: it tempers the inflation response while roughly doubling the output-gap response, and, through the risk-sharing condition and depreciation identity, the exchange rate inherits this enlarged adjustment—the initial response of nominal depreciation closely mirrors $\Gamma_{x,u}$. The signature of $\tilde{\theta}$ in the data is therefore a distinctive *conditional comovement*: episodes driven by cost-push shocks in which inflation is contained while the exchange rate adjusts substantially. This is precisely the pattern reflected in the variance decompositions of Table 5, where the NRE model attributes the bulk of exchange-rate fluctuations to cost-push shocks while the RE model attributes them to productivity.

The backward-looking fraction of firms δ can generate inflation persistence directly through the aggregate supply relation; but it operates only there and leaves the open-economy relations untouched, so it cannot generate the reallocation pattern itself—and, as Table 5 shows, even with δ freely estimated, the RE model attributes exchange-rate fluctuations to productivity rather than cost-push shocks. In the closed economy, the second observable imposes no comparable discipline. The closed-economy estimation absorbs the observed inflation dynamics through δ near one together with a small cost-push volatility, and the bulk of output-growth variation (over 80 percent) reflects movements in natural output, which are invariant to the policy and hence to $\tilde{\theta}$. Accordingly, the closed-economy NRE and RE models imply nearly identical variance decompositions for

Table 6: The Role of Openness: Closed- versus Open-Economy Estimates

	Closed economy		Open economy	
	NRE	RE	NRE	RE
<i>Panel A: Baseline</i>				
$\tilde{\theta}$: mean	0.076	–	1807	–
$\tilde{\theta}$: [5%, 95%]	[0.004, 0.233]	–	[896, 3190]	–
δ : mean	0.96	0.97	0.10	0.49
Log MDD	253.21	265.37	209.57	192.07
Policy-rate RMSE	0.0405	0.0406	0.0295	0.0788
<i>Panel B: No backward-looking firms</i>				
$\tilde{\theta}$: mean	34.33	–	1594	–
$\tilde{\theta}$: [5%, 95%]	[2.42, 98.30]	–	[831, 2708]	–
Log MDD	160.68	166.79	212.82	192.00
Policy-rate RMSE	0.1524	0.1550	0.0301	0.0910

Note. The closed-economy specifications impose $\alpha = 0$ and use CPI inflation and output growth as observables, with priors identical to the baseline estimation; details of the specification are provided in Appendix C. The open-economy columns reproduce the baseline results (Table 1). Because the closed- and open-economy estimations use different observable sets, log marginal data densities are comparable *within* an economy (across NRE and RE) but not *across* economies. The policy-rate RMSE is computed for the demeaned, annualized policy rate, as in Figure 3.

both observables¹⁸—in sharp contrast to the open economy, where the two belief structures imply starkly different attributions of exchange-rate fluctuations (Table 5). With the two models observationally near-equivalent on $\{\pi_t, \Delta y_t\}$, the closed-economy data offer no margin along which the belief distortions could improve the model’s account: the posterior of $\tilde{\theta}$ is concentrated near zero, and the concern for robustness is estimated to be absent.

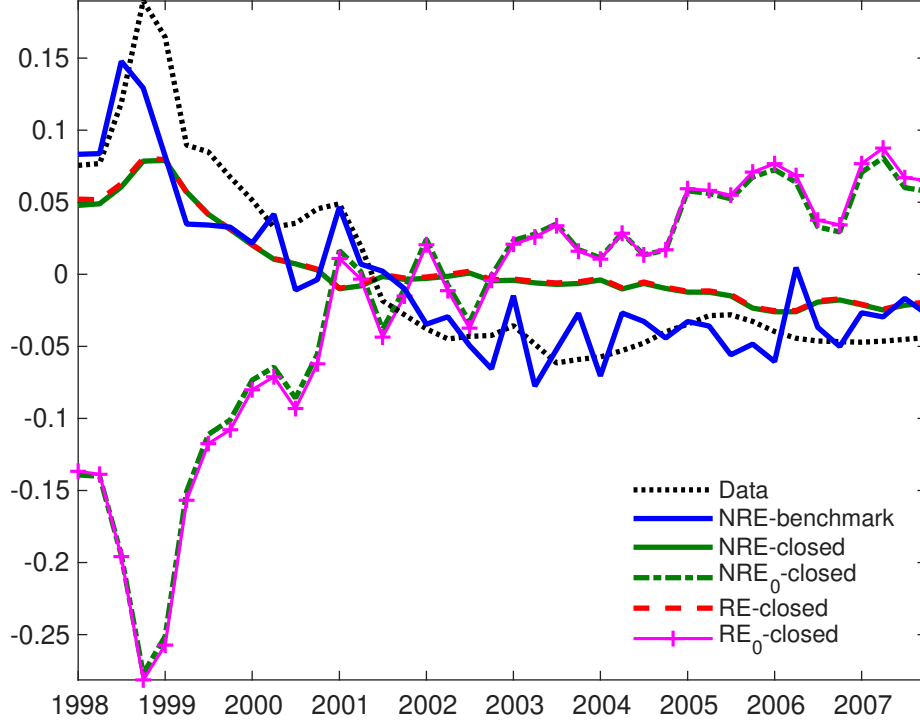
4 Discussion

4.1 Robustly Optimal Policy with Rational Private Beliefs

The baseline NRE model estimated in Section 3 embeds a self-confirming belief structure: the central bank designs policy to guard against a worst-case distortion of private-sector expectations, and private agents’ actual expectations coincide with that same worst-case distortion. In the notation introduced in Section 3 regarding the different probability models, objects (b) and (c) coincide. We now consider the polar opposite belief structure, in which private agents actually hold RE, so that the central bank’s concern for robustness is, in a precise sense, unfounded. We refer to this alternative specification as the private-sector rational expectations (PRE) model, to distinguish it from the baseline NRE specification in which private-sector beliefs coincide with the central bank’s worst case.

¹⁸In the baseline closed-economy estimation, cost-push shocks account for 97.7 (98.8) percent of inflation variance under NRE (RE), with productivity shocks contributing zero to inflation—the closed-economy counterpart of Lemma 3.1; the $\delta = 0$ variants display the same pattern.

Figure 6: Monetary Policy Rates: Data and Model Predictions: Closed Economy Models



Note. The black dotted line denotes the monetary policy rate in the data. The blue solid line (labeled NRE-benchmark) reproduces the baseline open-economy NRE prediction from Figure 3. The green solid line (labeled NRE-closed) and the red dashed line (labeled RE-closed) show the policy rate paths predicted by the estimated closed-economy NRE and RE models, respectively. The green dash-dotted line (labeled NRE₀-closed) and the magenta line with crosses (labeled RE₀-closed) show the predicted policy rate paths from the estimated closed-economy NRE and RE models without backward-looking firms (i.e., $\delta = 0$), respectively. All series are demeaned and annualized. Within each closed-economy pair, the NRE and RE series are nearly indistinguishable, reflecting the collapse of the identified $\hat{\theta}$ in the closed economy. The RMSEs for NRE-closed, RE-closed, NRE₀-closed, and RE₀-closed are 0.0405, 0.0406, 0.1524, and 0.1550, respectively.

Comparing NRE against PRE therefore lets us test, directly against the data, whether private agents' beliefs during the post-crisis transition are better described as coinciding with the worst-case distortion the central bank guards against, or as rational, and hence whether the central bank's robust policy stance was warranted by the beliefs that private agents actually held. We do so by comparing the estimation results across the models and contrasting how each model's implied policy-rate path tracks the data. Moreover, Section 4.4 complements this comparison with a formal detection error probability exercise.

The PRE model modifies the baseline NRE model in exactly one place. The central bank continues to set the nominal interest rate according to the robustly optimal reaction function derived from the min-max problem in Definition 2.1, together with the associated law of motion for the co-state Φ_t .¹⁹ Private agents, however, form expectations rationally: the aggregate demand

¹⁹The central bank does not observe which belief structure is operative and does not re-solve its optimization problem; its policy rule and the evolution of Φ_t therefore carry over unchanged.

Table 7: The Role of Private Beliefs: PRE versus NRE Estimates

	PRE	NRE	RE
<i>Panel A: Baseline</i>			
$\hat{\theta}$: mean	341.76	1807	—
$\tilde{\theta}$: [5%, 95%]	[189.10, 560.90]	[896, 3190]	—
δ : mean	0.10	0.10	0.49
Log MDD	208.19	209.57	192.07
Policy-rate RMSE	0.0758	0.0295	0.0788
<i>Panel B: No backward-looking firms</i>			
$\hat{\theta}$: mean	327.09	1594	—
$\tilde{\theta}$: [5%, 95%]	[137.12, 581.43]	[831, 2708]	—
Log MDD	208.29	212.82	192.00
Policy-rate RMSE	0.0798	0.0301	0.0910

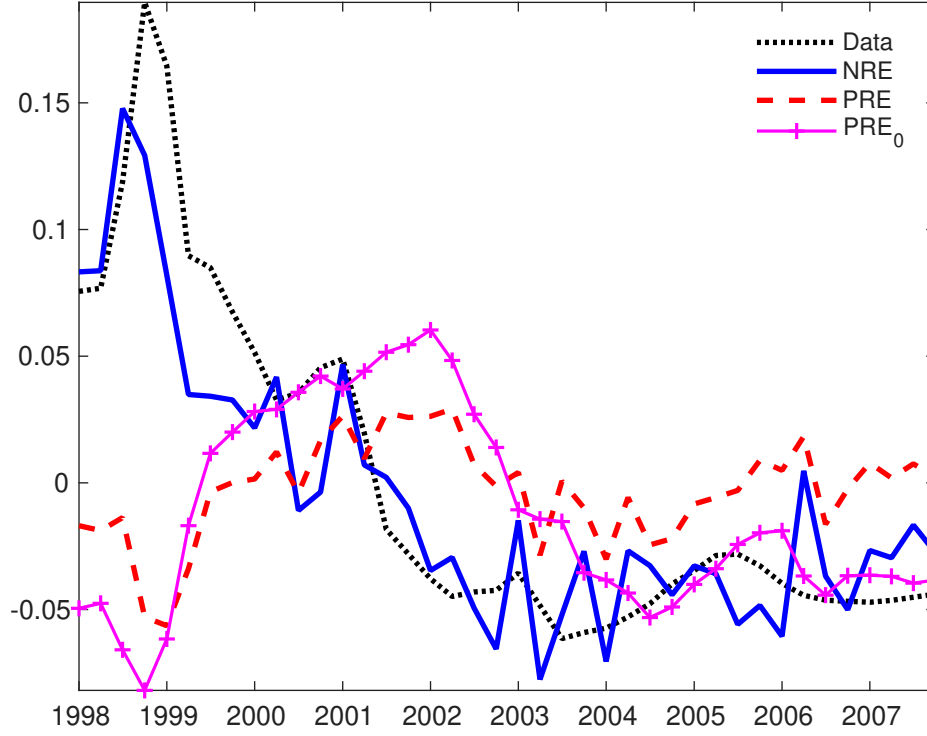
Note. The PRE specification retains the central bank’s robust rule but endows private agents with rational expectations, so that $\tilde{\theta}$ enters only through the policy rule; it is re-estimated freely with priors identical to the baseline estimation, and details of the specification are provided in Appendix D. The NRE and RE columns reproduce the baseline results (Table 1). The policy-rate RMSE is computed for the demeaned, annualized policy rate, as in Figure 3. Full posterior distributions of the PRE specifications are reported in Table E.12 in Appendix E.

and aggregate supply relations, (2.7) and (2.10), hold with $\mathbb{E}_t[\cdot]$ in place of $\hat{\mathbb{E}}_t[\cdot]$. We estimate the resulting model with the same Bayesian procedure, observables $(\pi_t, \Delta e_t)$, and priors as in Section 3; the measurement and transition equations are detailed in Appendix D.

Panel A of Table 7 summarizes the baseline results. The most consequential difference lies in the estimated concern for robustness itself: the posterior mean of $\tilde{\theta}$ falls from 1807 in the NRE model to 341.76—roughly one-fifth of the NRE estimate—once private agents hold rational beliefs. The strong concern for robustness identified in the NRE model thus rests on the belief distortions that private agents actually hold; absent these distortions, the data assign the policy rule only a modest concern for robustness. In terms of in-sample fit, the PRE model attains a log marginal data density of 208.19, within 1.4 log points of the NRE benchmark: conditional on rational private beliefs, the robustly optimal policy reproduces most of the in-sample dynamics of inflation and depreciation. Panel B shows that these results are robust to removing the backward-looking component altogether: $\tilde{\theta}$ again falls to roughly one-fifth of the NRE₀ estimate (327.09 against 1594), and the log marginal data density of PRE₀ remains within 4.6 log points of NRE₀.

Turning to the policy-rate dimension, Figure 7 plots the policy-rate paths implied by the NRE, PRE, and PRE₀ models against the actual data. On this dimension the rational-expectations alternatives fail alike in both panels: the RMSE is 0.0758 for PRE and 0.0798 for PRE₀, against 0.0295 and 0.0301 for their NRE counterparts, and the RE model, at 0.0788, fares no better. The evidence that separates the NRE model from its rational-expectations alternatives is thus not the in-sample fit—which the PRE models can largely replicate—but the ability to rationalize the central bank’s observed policy behavior. The two pieces of evidence are complementary: the estimation, based on the joint dynamics of observables, favors specifications with the robustly optimal policy

Figure 7: Monetary Policy Rates: Data and Model Predictions: PRE Models



Note. The black dotted line denotes the monetary policy rate in the data. The blue solid line (labeled NRE) reproduces the baseline NRE prediction from Figure 3. The red dashed line (labeled PRE) and the magenta line with crosses (labeled PRE_0) show the policy rate paths predicted by the estimated PRE models of Section 4.1, in which the central bank's robustly optimal rule is unchanged but private agents actually hold rational expectations, with and without backward-looking firms (i.e., $\delta = 0$), respectively. All series are demeaned and annualized. The RMSEs for NRE, PRE, and PRE_0 are 0.0295, 0.0758, and 0.0798, respectively.

over the RE model, while the policy-rate comparison discriminates between the two belief structures that fit the estimation sample comparably well.

4.2 Splitting the Sample: Pre- and Post-Inflation-Targeting Periods

To test whether the identified concern for robustness in the NRE model is indeed concentrated where the mechanism locates it—in periods of elevated cost-push volatility—we split the Mexican sample at the formal adoption of the inflation-targeting regime in 2001: the early sub-sample (1Q 1998–4Q 2001) covers the turbulent transition years through the adoption of the regime, and the later sub-sample (1Q 2002–4Q 2007) covers the period thereafter. We re-estimate the baseline NRE and RE models on each sub-sample, with priors and procedure identical to the full-sample estimation; full posterior distributions are reported in Table E.15 in Appendix E.

The results deliver three findings. First, the estimated cost-push volatility declines across the regime boundary, from $\sigma_u = 0.042$ in the early sub-sample to 0.028 in the later one. Second, the identified concern for robustness declines in step, though it does not vanish: the posterior mean

of $\tilde{\theta}$ falls from 3970 (90% credible interval [1738, 7373]) to 1098 ([159, 2585]). Third—and most tellingly—the empirical bite of the concern for robustness is concentrated in the early period. In the pre-adoption sub-sample, the NRE model decisively dominates the RE model: its log marginal data density is higher by 9.4 log points (78.41 versus 68.99), its policy-rate RMSE is less than half of the RE model’s (0.0443 versus 0.1028), and the RE model again requires a large backward-looking share to fit the data ($\delta = 0.51$ versus 0.17 under NRE). In the post-adoption sub-sample, by contrast, the data no longer meaningfully separate the two models: the marginal data densities are essentially tied (145.20 versus 146.72), the policy-rate RMSEs are nearly identical (0.0529 versus 0.0549), and neither model relies on backward-looking pricing ($\delta = 0.12$ under NRE and 0.07 under RE).

We read this evidence as follows. The concern for robustness is not a fixed trait of the central bank: both its magnitude and—especially—its empirical relevance co-move with the volatility environment, being largest in the transition years when the anchoring of expectations was most in doubt. The finding that $\tilde{\theta}$ declines without vanishing, while the NRE and RE models become indistinguishable, is itself informative about the mechanism: the worst-case belief distortion that policy guards against scales with the volatility of the underlying disturbances, so that in a quiet environment even a sizable concern for robustness leaves only a faint imprint on equilibrium dynamics—and the data lose the power to discriminate it from the RE model.

We note one caveat: the two sub-samples contain only 16 and 24 quarterly observations, respectively, so the posterior credible intervals are correspondingly wide, and we interpret the sub-sample estimates as indicative of the direction of change rather than as precise magnitudes.

4.3 Advanced Economy Counterpart

We conduct the same exercise as in Section 3 for an advanced economy, where we presume that the central bank faced significantly lower levels of Knightian uncertainty due to its long-standing experience with policy commitment and well-anchored inflation expectations, as well as the absence of abrupt policy regime shifts such as those observed in Mexico. Using Canadian data (2Q 1991 to 4Q 2007), we obtain results consistent with this presumption. The identified concern for robustness is more than an order of magnitude smaller than for Mexico ($\tilde{\theta} = 49.5$ and 37.4 with and without backward-looking pricing, versus 1807 and 1594), the estimated backward-looking share is negligible ($\delta = 0.06$), and the NRE and RE models yield similar parameter estimates. The corresponding results are reported in Table E.16 and Figure E.9 in Appendix E.

Two additional observations sharpen this contrast. First, the adequacy of the RE approximation in Canada is quantitative, not merely relative. The NRE model’s decisive in-sample advantage for Mexico over RE vanishes entirely in Canada, where the ranking in fact tilts slightly toward the RE model (416.23 for NRE versus 419.34 for RE, and similarly without backward-looking pricing). On external validity, the RE model’s policy-rate RMSE for Canada (0.0398) is 1.3 times the NRE model’s (0.0305), compared with 2.6 times for Mexico (0.0788 versus 0.0295): the RE

model approximates observed Canadian policy behavior about twice as well as it does Mexican policy behavior. Second, the estimated cost-push innovation volatility in Canada ($\sigma_u = 0.026$) is comparable to Mexico’s (0.030), so the size of the shocks alone cannot account for the thirty-six-fold difference in the identified concern for robustness. The two economies differ instead in where the estimates locate the persistence of inflation: the Canadian estimates assign it to the exogenous process itself ($\rho_u = 0.92$, versus 0.32 for Mexico), leaving neither the belief channel nor backward-looking pricing a separate role to play.

4.4 Detection Error Probability

The raw value of $\tilde{\theta}$ scales with the units of the loss function and is therefore not directly comparable across studies; the detection error probability (DEP) provides a natural metric for the size of the estimated model doubts. Following Barillas et al. (2009), we assign prior probability one-half each to the approximating model (model A) and to the worst-case model associated with a given $\tilde{\theta}$ (model B). After observing T periods of data, a likelihood-ratio test is performed: if model A were correct, the likelihood ratio would falsely attribute the data to model B a fraction p_A of the time, and conversely for p_B . The detection error probability is $p(\tilde{\theta}) = \frac{1}{2}(p_A + p_B)$. When $\tilde{\theta} = 0$ the two models coincide and $p = 0.5$; as $\tilde{\theta}$ rises above zero, $p(\tilde{\theta})$ falls below 0.5.

In our environment, this definition pins down the model pair. The worst-case model is the NRE equilibrium; the approximating model is its PRE counterpart of Section 4.1, which shares the central bank’s committed rule and the parameter vector—both evaluated at the NRE posterior—and differs only in the beliefs that private agents actually hold. As $\tilde{\theta} \rightarrow 0$ the two equilibria coincide and $p \rightarrow 0.5$, as the definition requires. Both equilibria are driven by undistorted innovations, since the type-3 distortion resides in the expectations rather than in the shocks, and the discriminating statistic is the likelihood ratio implied by the central bank’s worst-case belief distortion. The resulting DEP therefore measures how much data the central bank would need to distinguish its worst-case fears from rational private beliefs, holding its committed rule fixed—so that the likelihood ratio detects the belief distortion and nothing else.²⁰

For Mexico, the DEP at the sample length of our data ($T = 40$) is 0.071 at the posterior mean, with a 90% credible interval of [0.024, 0.106]. Although $\tilde{\theta}$ is estimated freely rather than calibrated to a detection-error target, the implied probability is thus close to the 10 percent level customarily targeted when calibrating the robustness parameter (e.g., Anderson et al., 2003; Maenhout, 2006; Kwon and Miao, 2017), and the credible interval contains it. Over the turbulent early portion of the sample—1998 through 2001, before full-fledged inflation targeting—the accumulated data reach $T = 16$ quarters, at which the DEP is 0.189, within the range that Barillas et al. (2009) characterize as “conservative” and consider it sensible to guard against (detection error probabilities of “0.2 or even less”). Figure E.11 in Appendix E plots the posterior distribution of the DEP as

²⁰We propagate the full joint posterior through this mapping: 1,000 joint posterior draws, with 10,000 likelihood-ratio simulations per draw.

a function of the sample length for both countries. The same computation for Canada yields 0.438 ([0.385, 0.476]) at its sample length of $T = 67$: in the tranquil advanced economy the two belief regimes are statistically nearly indistinguishable—the detection-error expression of the small concern for robustness implied by the Canadian estimates.

4.5 Sensitivity Checks of the Main Results

We re-estimate the NRE and NRE₀ models in Section 3 by imposing a wider prior for $\tilde{\theta}$, specifically uniform distributions over the ranges $[0, 10^6]$ and $[0, 10^{18}]$. The corresponding posterior distributions for the NRE models are reported in Table E.13 in Appendix E. We find no statistically meaningful differences in the posterior distributions of the NRE and NRE₀ models compared to those in Table 1, and the main results in Section 3 remain unchanged.

We also repeat the exercise in Section 3 using extended samples that begin closer to the Mexican Peso Crisis, starting in 1Q 1996 and 1Q 1997. The results remain robust: the NRE models exhibit a substantial degree of concern for robustness (high values of $\tilde{\theta}$), rely only minimally on the backward-looking term in the aggregate supply, and deliver strong external-validity performance in predicting the monetary policy rate. By contrast, the RE models rely heavily on the backward-looking term and still fall short in predictive performance. The corresponding results are reported in Table E.14 and Figure E.8 in Appendix E.

We also conduct a sensitivity analysis with respect to the pre-set parameters: the subjective discount factor β and the Calvo–Yun parameter ζ . We assess their role by re-estimating all four specifications (NRE, RE, NRE₀, RE₀) over a grid of nine (β, ζ) combinations. For the discount factor, the grid spans $\beta \in \{0.965, 0.975, 0.99\}$, from the lower values typical of emerging-market calibrations to the standard advanced-economy value of 0.99. For the Calvo–Yun parameter, the grid spans $\zeta \in \{0.75, 0.50, 0.25\}$, from the baseline duration of one year down to roughly 1.3 quarters. The lowest value is motivated by micro evidence: using Mexican CPI micro data for 1994–2002, Gagnon (2009) documents that roughly a third of consumer goods prices changed each month toward the end of the period.

Table E.17 summarizes the results. Every main finding of the paper is preserved throughout the grid. The estimated concern for robustness remains large, with posterior means of $\tilde{\theta}$ between 1807 and 2196 (between 1557 and 1910 for NRE₀); the backward-looking share δ remains small under NRE (0.07–0.11), while the RE model continues to require δ around one-half (0.48–0.58); the marginal data density favors NRE over RE by 11.0 to 17.5 log points (20.1 to 24.1 for the specifications without backward-looking price setting); and the policy-rate RMSE separation survives, with 0.0276–0.0298 for NRE against 0.0574–0.0794 for RE.

4.6 Financial Frictions and Expectation-Based Mechanisms

Our findings also inform the broader debate on the role of financial frictions in emerging-market monetary policy. While a large literature has emphasized the role of monetary policy in the presence of various forms of financial frictions—including sovereign default risk (e.g., [Na et al., 2018](#)), sudden stops and external financing constraints (e.g., [Braggion et al., 2009](#); [Devereux et al., 2019](#)), and currency risk premia (e.g., [Itskhoki and Mukhin, 2023](#))—our results indicate that even in the absence of such wedges, belief distortions combined with policymakers’ concern for robustness can generate quantitatively meaningful policy dynamics.

Importantly, this does not imply that financial frictions are unimportant. Rather, it suggests that part of what is often attributed to financial wedges may, in certain environments, reflect uncertainty about private-sector expectation formation. For policy analysis, this distinction matters: it shifts the focus from correcting external financing distortions to stabilizing expectations and managing robustness concerns within the domestic policy framework.

An open question is whether and how ambiguity aversion interacts with financial frictions. Our sample speaks to the substitution margin by design: although the early transition years feature elevated cost-push volatility and a correspondingly high identified concern for robustness (Section 4.2), Mexico did not experience a sudden stop, a sovereign default episode, or a binding external financing constraint over 1998–2007, so frictions of the kind emphasized in that literature were unlikely to be first-order. Whereas in episodes with binding financial frictions, worst-case belief distortions could act either as a substitute for or as a complement to such frictions. A formal treatment would require embedding financial frictions into the robust policy problem, which we leave for future research.

5 Conclusion

This paper studies robustly optimal monetary policy in a small open economy when the central bank faces uncertainty about private agents’ forward-looking expectations. We introduce near-rational expectations (NRE) into an otherwise standard open-economy New Keynesian framework and model the policymaker’s concern for robustness as ambiguity about belief distortions. Within this environment, optimal policy departs sharply from the rational-expectations (RE) benchmark: greater concern for robustness leads the robustly optimal policy to keep domestic inflation more contained in the presence of cost-push shocks, at the cost of tolerating larger adjustments in the output gap and the nominal exchange rate. Moreover, the robustly optimal monetary policy exhibits endogenous history dependence and generates persistent inflation dynamics even in the absence of backward-looking price setting.

We take the model to the data by estimating both NRE and RE versions using Mexican macroeconomic time series during the post–Peso Crisis transition. The estimated NRE model implies a substantial degree of concern for robustness on the part of the central bank and delivers a markedly

different policy configuration from the RE benchmark. In particular, the NRE model assigns a central role to the interaction between a concern for robustness and cost-push shocks in shaping inflation, exchange-rate movements, and policy-rate dynamics, whereas the RE model relies heavily on backward-looking pricing behavior and productivity shocks. These differences are not merely quantitative: the NRE framework substantially outperforms the RE benchmark in external validity, closely tracking the observed path of Mexico’s policy interest rate despite not using it as an observable in estimation.

The interpretation of the identified concern for robustness is corroborated from several directions. It is identified only when the open-economy structure and the exchange-rate observable are present; it is concentrated, in both magnitude and empirical relevance, in the early transition years before the inflation-targeting regime was adopted; and it cannot be imitated by rational-expectations alternatives, which fail, with or without the robust rule, to rationalize the observed monetary policy path in Mexico.

Beyond its empirical performance, the analysis highlights a novel policy mechanism with broader relevance for emerging market economies. When central banks face uncertainty about the formation of private-sector expectations—particularly in the aftermath of crisis-driven regime transitions—robust policy design reshapes both shock transmission and the interpretation of inflation persistence. In our framework, inflation inertia emerges endogenously from optimal policy responses under ambiguity, rather than from mechanical backward-looking behavior. This distinction suggests that observed inflation persistence in post-crisis environments need not reflect intrinsic structural rigidities, but may instead arise from deliberate policy choices aimed at guarding against worst-case distortions in expectations.

Taken together, our results underscore the importance of incorporating policymakers’ uncertainty about private-sector expectations into open-economy monetary models. Even within a parsimonious structural framework, accounting for robustness concerns yields qualitatively different policy prescriptions and empirical implications. We view this approach as a promising avenue for understanding monetary policy design in economies undergoing institutional transitions—in Mexico and beyond—where anchoring private-sector expectations remains a central—and inherently uncertain—policy challenge.

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Monetary Policy under Uncertain Expectations in an Emerging Economy

Appendix: For Online Publication

A Details in Section 2

A.1 The Aggregate Supply with Lagged Domestic Inflation

The households' log-linearized optimal choices of consumption and labor are the same as those in [Galí and Monacelli \(2005\)](#) under distorted beliefs:

$$\sigma c_t + \varphi n_t = w_t - p_t, \quad (\text{A.1})$$

$$c_t = \hat{\mathbb{E}}_t[c_{t+1}] - \frac{1}{\sigma} \left(r_t - \hat{\mathbb{E}}_t[\pi_{t+1}] - \rho \right). \quad (\text{A.2})$$

The firms also have the same production technology as the ones in [Galí and Monacelli \(2005\)](#): $Y_t(j) = A_t N_t(j)$. Now, following [Galí and Gertler \(1999\)](#), we assume that a fraction $1 - \delta$ of the firms, which we refer to as forward-looking, behave like the firms in Calvo–Yun model, i.e., they set prices optimally, given the constraints on the timing of adjustments and using all the available information in order to forecast future marginal costs. The remaining firms of measure δ , which we refer to as backward-looking, instead use a simple rule of thumb that is based on the recent history of aggregate price behavior.

Every period, a fraction $1 - \zeta$ of firms reset prices (as in Calvo–Yun):

$$P_{H,t} = \left[\zeta (P_{H,t-1})^{1-\epsilon} + (1 - \zeta) (\bar{P}_{H,t}^{reset})^{1-\epsilon} \right]^{\frac{1}{1-\epsilon}},$$

and out of those that reset, some are forward-looking ($1 - \delta$) while others are backward-looking (δ):

$$\bar{P}_{H,t}^{reset} = \left[(1 - \delta) (P_{H,t}^f)^{1-\epsilon} + \delta (P_{H,t}^b)^{1-\epsilon} \right]^{\frac{1}{1-\epsilon}},$$

where the backward-looking firms set:

$$P_{H,t}^b = \bar{P}_{H,t-1}^{reset} \times \Pi_{H,t-1},$$

and the forward-looking firms solve the optimal pricing problem:

$$\max_{P_{H,t}^f} \sum_{k=0}^{\infty} \zeta^k \hat{\mathbb{E}}_t \left\{ Q_{t,t+k} [P_{H,t}^f Y_{t+k}(j) - (1 - \tau_f) W_{t+k} Y_{t+k}(j) / A_{t+k}] \right\},$$

subject to

$$Y_{t+k}(j) = \left(\frac{P_{H,t}^f}{P_{H,t+k}} \right)^{-\epsilon} \left(C_{H,t+k} + \int_0^1 C_{H,t+k}^i di \right),$$

where $Q_{t,t+k}$ is the households' one-period stochastic discount factor and τ_f is the employment subsidy. The firms' optimization yields the following log-linearized equation:

$$p_{H,t}^f = (1 - \beta\zeta)(\mu + mc_t + p_{H,t}) + \beta\zeta \hat{\mathbb{E}}_t [p_{H,t+1}^f].$$

The aggregate price level can, therefore, be solved for using the solutions of $p_{H,t}^f$ and $p_{H,t}^b$ in the expression of $\bar{P}_{H,t}^{reset}$. This price level can then be used to produce the following log-linearized equation:

$$\pi_{H,t} = \frac{(1 - \zeta)(1 - \delta)(1 - \beta\zeta)}{[\zeta + \delta(1 - \zeta(1 - \beta))]}(\mu + mc_t) + \frac{\delta}{[\zeta + \delta(1 - \zeta(1 - \beta))]} \pi_{H,t-1} + \frac{\beta\zeta}{[\zeta + \delta(1 - \zeta(1 - \beta))]} \hat{\mathbb{E}}_t \pi_{H,t+1},$$

where the coefficients are written as:

$$\tilde{\beta} \equiv \frac{\zeta}{[\zeta + \delta(1 - \zeta(1 - \beta))]} \cdot \beta < \beta,$$

$$\eta \equiv \frac{\delta}{[\zeta + \delta(1 - \zeta(1 - \beta))]} > 0.$$

Finally, using the expression for marginal cost, $\mu + mc_t = (\tau + \varphi)x_t$, the aggregate supply curve augmented with the lagged domestic inflation is given as:

$$\pi_{H,t} = \tilde{\beta} \hat{\mathbb{E}}_t \{\pi_{H,t+1}\} + \eta \pi_{H,t-1} + \kappa x_t,$$

where

$$\kappa \equiv \frac{\zeta(1 - \delta)\kappa_0}{[\zeta + \delta(1 - \zeta(1 - \beta))]} < \kappa_0,$$

consistent with the definitions in (2.11). Adding the exogenous cost-push disturbance u_t delivers equation (2.10) in the main text.

A.2 Solving the LQ System with Conditionally Linear Commitment

This section describes how we solve the linear-quadratic (LQ) robust policy problem under conditionally linear commitment, following [Kwon and Miao \(2019\)](#). The solution proceeds in the following steps. First, a hypothetical malevolent agent chooses the worst-case distortion to maximize welfare losses, subject to a relative-entropy penalty. Second, the central bank chooses the allocation and the policy instrument, taking the worst-case distortion as given. These two problems jointly yield a stacked linear system, from which we obtain the state-space representation by applying a standard linear difference equation solver. Since the conditionally linear self-commitment

is time-invariant, we update the solution iteratively until a specified convergence criterion is met.

Worst-case beliefs. The first-order condition of the Lagrangian (2.21) with respect to m_{t+1} is:

$$\tilde{\theta}^{-1}(1 + \ln m_{t+1}) - \phi_t - \boldsymbol{\mu}'_{Y,t} \mathbf{Y}_{t+1} = 0.$$

Using the representation $\mathbf{Y}_{t+1} = \boldsymbol{\Phi}_t + \boldsymbol{\Gamma}_t \boldsymbol{\varepsilon}_{t+1}$ for $t \geq 0$ and the normalization constraint (2.1), we obtain

$$m_{t+1} = \exp \left(-\frac{1}{2} \tilde{\theta}^2 \boldsymbol{\mu}'_{Y,t} \boldsymbol{\Gamma}_t \boldsymbol{\Gamma}'_t \boldsymbol{\mu}_{Y,t} + \tilde{\theta} \boldsymbol{\mu}'_{Y,t} \boldsymbol{\Gamma}_t \boldsymbol{\varepsilon}_{t+1} \right).$$

This implies the following conditional moments under the worst-case beliefs:

$$\begin{aligned} \mathbb{E}_t m_{t+1} \mathbf{Y}_{t+1} &= \boldsymbol{\Phi}_t + \tilde{\theta} \boldsymbol{\Gamma}_t \boldsymbol{\Gamma}'_t \boldsymbol{\mu}_{Y,t}, \\ \mathbb{E}_t m_{t+1} \ln m_{t+1} &= \frac{1}{2} \tilde{\theta}^2 \boldsymbol{\mu}'_{Y,t} \boldsymbol{\Gamma}_t \boldsymbol{\Gamma}'_t \boldsymbol{\mu}_{Y,t}. \end{aligned}$$

Central bank's problem and first-order conditions. Substituting the worst-case distortion into the objective, the period loss can be written in matrix form as

$$\begin{aligned} L(\mathbf{X}_t, \boldsymbol{\Phi}_{t-1} + \boldsymbol{\Gamma}_{t-1} \boldsymbol{\varepsilon}_t, i_t) &= \frac{1}{2} \begin{bmatrix} \mathbf{X}_t \\ \boldsymbol{\Phi}_{t-1} + \boldsymbol{\Gamma}_{t-1} \boldsymbol{\varepsilon}_t \end{bmatrix}' \mathbf{Q} \begin{bmatrix} \mathbf{X}_t \\ \boldsymbol{\Phi}_{t-1} + \boldsymbol{\Gamma}_{t-1} \boldsymbol{\varepsilon}_t \end{bmatrix} \\ &\quad + \frac{1}{2} i_t \mathbf{R} i_t + \begin{bmatrix} \mathbf{X}_t \\ \boldsymbol{\Phi}_{t-1} + \boldsymbol{\Gamma}_{t-1} \boldsymbol{\varepsilon}_t \end{bmatrix}' \mathbf{S} i_t, \end{aligned}$$

where the matrix $\begin{bmatrix} \mathbf{Q} & \mathbf{S} \\ \mathbf{S}' & \mathbf{R} \end{bmatrix}$ is symmetric and positive definite.

The central bank chooses $\{\mathbf{X}_t, \boldsymbol{\Phi}_t, \boldsymbol{\Gamma}_t, i_t\}$ after substituting for the chosen value of m_{t+1} in the Lagrangian. The new Lagrangian is:

$$\begin{aligned} \mathcal{L} &= \mathbb{E}_{-1} \sum_{t=0}^{\infty} \beta^t \left\{ L(\mathbf{X}_t, \boldsymbol{\Phi}_{t-1} + \boldsymbol{\Gamma}_{t-1} \boldsymbol{\varepsilon}_t, i_t) - \frac{\tilde{\theta}}{2} \boldsymbol{\mu}'_{Y,t} \boldsymbol{\Gamma}_t \boldsymbol{\Gamma}'_t \boldsymbol{\mu}_{Y,t} + \right. \\ &\quad \left. \begin{bmatrix} \boldsymbol{\mu}_{X,t+1} \\ \boldsymbol{\mu}_{Y,t} \end{bmatrix}' \left(\begin{bmatrix} \mathbf{X}_{t+1} \\ \boldsymbol{\Phi}_t + \tilde{\theta} \boldsymbol{\Gamma}_t \boldsymbol{\Gamma}'_t \boldsymbol{\mu}_{Y,t} \end{bmatrix} - \mathbf{A} \begin{bmatrix} \mathbf{X}_t \\ \boldsymbol{\Phi}_{t-1} + \boldsymbol{\Gamma}_{t-1} \boldsymbol{\varepsilon}_t \end{bmatrix} - \mathbf{B} i_t - \mathbf{C} \boldsymbol{\varepsilon}_{t+1} \right) \right\}. \end{aligned}$$

The first-order necessary conditions with respect to $\{\mathbf{X}_t, \boldsymbol{\Phi}_t, \boldsymbol{\Gamma}_t, i_t\}$ are:

$$\begin{aligned} \mathbf{0} &= -\mathbf{Q}_{XX} \mathbf{X}_t - \mathbf{Q}_{XY} (\boldsymbol{\Phi}_{t-1} + \boldsymbol{\Gamma}_{t-1} \boldsymbol{\varepsilon}_t) - \mathbf{S}_X i_t - \beta^{-1} \boldsymbol{\mu}_{X,t} + \mathbf{A}'_{XX} \mathbb{E}_t \boldsymbol{\mu}_{X,t+1} + \mathbf{A}'_{YX} \boldsymbol{\mu}_{Y,t}, \\ \mathbf{0} &= -\mathbf{R} i_t - \mathbf{S}'_X \mathbf{X}_t - \mathbf{S}'_Y (\boldsymbol{\Phi}_{t-1} + \boldsymbol{\Gamma}_{t-1} \boldsymbol{\varepsilon}_t) + \mathbf{B}'_X \mathbb{E}_t \boldsymbol{\mu}_{X,t+1} + \mathbf{B}'_Y \boldsymbol{\mu}_{Y,t}, \\ \mathbf{0} &= -(\mathbf{Q}_{YY} \boldsymbol{\Phi}_t + \mathbf{Q}'_{XY} \mathbb{E}_t \mathbf{X}_{t+1}) - \mathbf{S}_Y \mathbb{E}_t i_{t+1} - \beta^{-1} \boldsymbol{\mu}_{Y,t} + \mathbb{E}_t [\mathbf{A}'_{XY} \boldsymbol{\mu}_{X,t+2} + \mathbf{A}'_{YY} \boldsymbol{\mu}_{Y,t+1}], \\ \mathbf{0} &= -\beta \mathbf{Q}_{YY} \mathbb{E}_t [\boldsymbol{\Phi}_t \boldsymbol{\varepsilon}'_{t+1} + \boldsymbol{\Gamma}_t \boldsymbol{\varepsilon}_{t+1} \boldsymbol{\varepsilon}'_{t+1}] - \beta \mathbf{Q}'_{XY} \mathbb{E}_t \mathbf{X}_{t+1} \boldsymbol{\varepsilon}'_{t+1} \end{aligned}$$

$$-\beta \mathbf{S}_Y \mathbb{E}_t i_{t+1} \boldsymbol{\varepsilon}_{t+1}' - \tilde{\theta} \boldsymbol{\mu}_{Y,t} \boldsymbol{\mu}_{Y,t}' \boldsymbol{\Gamma}_t + \beta \mathbb{E}_t \left[\left(\mathbf{A}_{XY}' \boldsymbol{\mu}_{X,t+2} + \mathbf{A}_{YY}' \boldsymbol{\mu}_{Y,t+1} \right) \boldsymbol{\varepsilon}_{t+1}' \right] \quad (\text{A.3})$$

where the matrices are partitioned as $\mathbf{A} \equiv \begin{bmatrix} \mathbf{A}_{XX} & \mathbf{A}_{XY} \\ \mathbf{A}_{YX} & \mathbf{A}_{YY} \end{bmatrix}$, $\mathbf{B} \equiv [\mathbf{B}_X, \mathbf{B}_Y]'$, and $\mathbf{C} \equiv [\mathbf{C}_X, \mathbf{C}_Y]'$,
 $\mathbf{Q} \equiv \begin{bmatrix} \mathbf{Q}_{XX} & \mathbf{Q}_{XY} \\ \mathbf{Q}_{XY}' & \mathbf{Q}_{YY} \end{bmatrix}$, and $\mathbf{S} \equiv \begin{bmatrix} \mathbf{S}_X \\ \mathbf{S}_Y \end{bmatrix}$.

In addition, the first-order necessary conditions with respect to $\{\boldsymbol{\mu}_{X,t+1}, \boldsymbol{\mu}_{Y,t}\}$ deliver the law of motion of the constraints:

$$\begin{aligned} \mathbf{0} &= \mathbf{X}_{t+1} - \mathbf{A}_{XX} \mathbf{X}_t - \mathbf{A}_{XY} (\boldsymbol{\Phi}_{t-1} + \boldsymbol{\Gamma}_{t-1} \boldsymbol{\varepsilon}_t) - \mathbf{B}_X i_t - \mathbf{C}_X \boldsymbol{\varepsilon}_{t+1}, \\ \mathbf{0} &= \boldsymbol{\Phi}_t + \tilde{\theta} \boldsymbol{\Gamma}_t \boldsymbol{\Gamma}_t' \boldsymbol{\mu}_{Y,t} - \mathbf{A}_{YX} \mathbf{X}_t - \mathbf{A}_{YY} (\boldsymbol{\Phi}_{t-1} + \boldsymbol{\Gamma}_{t-1} \boldsymbol{\varepsilon}_t) - \mathbf{B}_Y i_t. \end{aligned}$$

Stacked linear system. Adding two auxiliary conditions $\mathbb{E}_t \boldsymbol{\varepsilon}_{t+1} = \mathbf{0}$ and $\mathbb{E}_t \boldsymbol{\mu}_{X,t+2} = \mathbb{E}_t \boldsymbol{\mu}_{X,t+1}$, the equilibrium conditions can be written as the stacked system

$$\mathbf{J} \begin{bmatrix} \mathbb{E}_t \boldsymbol{\varepsilon}_{t+1} \\ \mathbb{E}_t \mathbf{X}_{t+1} \\ \boldsymbol{\Phi}_t \\ \mathbb{E}_t i_{t+1} \\ \mathbb{E}_t \boldsymbol{\mu}_{Y,t+1} \\ \mathbb{E}_t \boldsymbol{\mu}_{X,t+1} \\ \mathbb{E}_t \boldsymbol{\mu}_{X,t+2} \end{bmatrix} = \mathbf{F} \begin{bmatrix} \boldsymbol{\varepsilon}_t \\ \mathbf{X}_t \\ \boldsymbol{\Phi}_{t-1} \\ i_t \\ \boldsymbol{\mu}_{Y,t} \\ \boldsymbol{\mu}_{X,t} \\ \mathbb{E}_t \boldsymbol{\mu}_{X,t+1} \end{bmatrix}, \quad (\text{A.4})$$

where $\{\boldsymbol{\varepsilon}_t, \mathbf{X}_t, \boldsymbol{\Phi}_{t-1}\}$ are predetermined state variables, whereas $\{i_t, \boldsymbol{\mu}_{Y,t}, \boldsymbol{\mu}_{X,t}, \mathbb{E}_t \boldsymbol{\mu}_{X,t+1}\}$ are non-predetermined variables. The explicit expressions for $\mathbf{J}$ and $\mathbf{F}$ follow [Kwon and Miao \(2017\)](#) and are reported below:

$$\mathbf{J} = \begin{pmatrix} \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{A}_{XX}' & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{B}_X' & \mathbf{0} \\ \mathbf{0} & \mathbf{Q}_{XY}' & \mathbf{Q}_{YY} & \mathbf{S}_Y & -\mathbf{A}_{YY}' & \mathbf{0} & -\mathbf{A}_{XY}' \end{pmatrix},$$

$$\mathbf{F} = \begin{pmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} \\ \mathbf{A}_{XY}\mathbf{\Gamma} & \mathbf{A}_{XX} & \mathbf{A}_{XY} & \mathbf{B}_X & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{A}_{YY}\mathbf{\Gamma} & \mathbf{A}_{YX} & \mathbf{A}_{YY} & \mathbf{B}_Y & -\tilde{\theta}\mathbf{\Gamma}\mathbf{\Gamma}' & \mathbf{0} & \mathbf{0} \\ \mathbf{Q}_{XY}\mathbf{\Gamma} & \mathbf{Q}_{XX} & \mathbf{Q}_{XY} & \mathbf{S}_X & -\mathbf{A}'_{YX} & \beta^{-1}\mathbf{I} & \mathbf{0} \\ \mathbf{S}'_Y\mathbf{\Gamma} & \mathbf{S}'_X & \mathbf{S}'_Y & \mathbf{R} & -\mathbf{B}'_Y & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & -\beta^{-1}\mathbf{I} & \mathbf{0} & \mathbf{0} \end{pmatrix}.$$

In our model, the partitions of the matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{S}$ are produced below:

$$\mathbf{A}_{XX} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \rho_u & 0 & 0 \\ 0 & 0 & 0 & \rho_a & 0 \\ 0 & 0 & 0 & 0 & \rho_{y^*} \end{bmatrix}, \quad \mathbf{A}_{XY} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix},$$

$$\mathbf{A}_{YX} = \begin{bmatrix} 0 & -\frac{\eta}{\beta} & -\frac{1}{\beta} & 0 & 0 \\ -\frac{\rho}{\tau} & \frac{\eta}{\tau\beta} & \frac{1}{\tau\beta} & \Lambda_{y,a}(1-\rho_a) & \alpha\left(\Theta + \frac{\Theta\tau}{\tau+\varphi}\right)(1-\rho_{y^*}) \end{bmatrix}, \quad \mathbf{A}_{YY} = \begin{bmatrix} \frac{1}{\beta} & -\frac{\tilde{\kappa}}{\beta} \\ -\frac{1}{\tau\beta} & 1 + \frac{\tilde{\kappa}}{\tau\beta} \end{bmatrix},$$

$$\mathbf{B}_X = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{B}_Y = \begin{bmatrix} 0 \\ \tau^{-1} \end{bmatrix}, \quad \mathbf{C}_X = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \sigma_u & 0 & 0 \\ 0 & \sigma_a & 0 \\ 0 & 0 & \sigma_{y^*} \end{bmatrix}, \quad \mathbf{C}_Y = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$\mathbf{Q}_{XX} = \begin{bmatrix} \lambda_x(\bar{x})^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{Q}_{XY} = \begin{bmatrix} 0 & -\lambda_x\bar{x} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{Q}_{YX} = \mathbf{Q}'_{XY}, \quad \mathbf{Q}_{YY} = \begin{bmatrix} 1 & 0 \\ 0 & \lambda_x \end{bmatrix},$$

$$\mathbf{R} = [0], \quad \mathbf{S}_X = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{S}_Y = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

We solve (A.4) using Klein (2000)'s method. The solution to the aforementioned system therefore takes the state-space representation as in (2.22) and (2.23).

Updating Γ . Under conditionally linear self-consistent commitment, Γ_t is time-invariant and is updated iteratively. Taking unconditional expectations in (A.3) and rearranging with respect to Γ yields the updating rule

$$\Gamma^{(n)} = \beta \left(\tilde{\theta} \mathbb{E} [\mu_{Y,t} \mu'_{Y,t}] + \beta Q_{YY} \right)^{-1} \left\{ \mathbb{E} [(A'_{XY} \mu_{X,t+2} + A'_{YY} \mu_{Y,t+1}) \varepsilon'_{t+1}] - Q_{XY} C_X - S_Y \mathbb{E} [i_t \varepsilon'_t] \right\}. \quad (\text{A.5})$$

To compute the moments appearing in (A.5), define the auxiliary state $\tilde{X}_t \equiv [\varepsilon'_t, X'_t, \Phi'_{t-1}]'$. From (2.22), the second moments satisfy

$$\mathbb{E} [\tilde{X}_{t+1} \tilde{X}'_{t+1}] = H [\tilde{X}_t \tilde{X}'_t] H' + \begin{pmatrix} I & C'_X & 0 \\ C_X & C_X C'_X & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

If the matrix H is stable, this Lyapunov equation yields $\mathbb{E} [\tilde{X}_t \tilde{X}'_t]$. After that, we apply (2.23) and derive

$$\mu_{Y,t} = G_{2,\varepsilon} \varepsilon_t + G_{2,X} X_t + G_{2,\Phi} \Phi_{t-1} = G_2 \tilde{X}_t,$$

where $G_2 \equiv [G_{2,\varepsilon}, G_{2,X}, G_{2,\Phi}]$ is the second row of G in (2.23). Then we obtain

$$\mathbb{E} [\mu_{Y,t} \mu'_{Y,t}] = G_2 \mathbb{E} [\tilde{X}_t \tilde{X}'_t] G'_2,$$

$$\mathbb{E} \begin{pmatrix} i_{t+1} \varepsilon'_{t+1} \\ \mu_{Y,t+1} \varepsilon'_{t+1} \\ \mu_{X,t+1} \varepsilon'_{t+1} \end{pmatrix} = G \mathbb{E} \begin{pmatrix} \varepsilon_{t+1} \varepsilon'_{t+1} \\ X_{t+1} \varepsilon'_{t+1} \\ \Phi_t \varepsilon'_{t+1} \end{pmatrix} = G \begin{pmatrix} I \\ C_X \\ 0 \end{pmatrix}.$$

The updating procedure of Γ in (A.5) stops when $\|\Gamma^{(n-1)} - \Gamma^{(n)}\| < \varepsilon_\Gamma$. We set $\varepsilon_\Gamma = 10^{-5}$.

A.3 Proofs

Proof of Lemma 3.1. The interest rate i_t is a choice variable of the central bank to determine the optimal path of domestic inflation $\pi_{H,t}$ and output gap x_t . At any given equilibrium path of $\{\pi_{H,t}, x_t\}$ and worst-case expectations, the corresponding path of interest rate makes the aggregate demand equation (2.7) always hold. Thus, the path of $\{\pi_{H,t}, x_t\}$ can be determined without reference to equation (2.7), as long as the zero lower bound is not binding, which is implicitly assumed in the model. Then the natural rate shocks $\{a_t, y_t^*\}$ in (2.7) do not affect the optimal paths of $\{\pi_{H,t}, x_t\}$. Thus, $\Gamma_{\pi_H, a} = \Gamma_{x, a} = \Gamma_{\pi_H, y^*} = \Gamma_{x, y^*} = 0$. $\square$

Proof of Proposition 3.2. Consider the case with $\delta = 0$. Given the worst-case near-rational expectations of private agents, the first-order condition of the central bank's minimization problem

in (2.19) with respect to $\Phi_{\pi_H,t}$ can be written as (see Appendix A.2 in [Woodford, 2010](#))

$$-\beta \frac{\lambda_x}{\kappa} \frac{\pi_{H,t} - u_t - \kappa \bar{x} - \beta \Phi_{\pi_H,t}}{\Xi} + \beta \mathbb{E}_t \left(\pi_{H,t+1} + \frac{\lambda_x}{\kappa^2} \frac{\pi_{H,t+1} - u_{t+1} - \kappa \bar{x} - \beta \Phi_{\pi_H,t+1}}{\Xi} \right) = 0, \quad (\text{A.6})$$

where

$$\Xi \equiv 1 - \tilde{\theta} \beta^2 \frac{\lambda_x}{\kappa^2} \Gamma_{\pi_H,u}^2 > 0.$$

Under the conditionally linear commitment policy, domestic inflation satisfies

$$\pi_{H,t} = \Phi_{\pi_H,t-1} + \Gamma_{\pi_H,u} \varepsilon_t^u.$$

Substituting this relation into (A.6) and rearranging terms yields the following second-order stochastic difference equation for $\Phi_{\pi_H,t}$:

$$\mathbb{E}_t[A(L)\Phi_{\pi_H,t+1}] = (\sigma_u - \Gamma_{\pi_H,u})\varepsilon_t^u - \rho_u(u_t - u_{t-1}), \quad (\text{A.7})$$

where L denotes the lag operator and

$$A(L) \equiv \beta - \left(1 + \beta + \frac{\kappa^2 \Xi}{\lambda_x} \right) L + L^2. \quad (\text{A.8})$$

Factorizing the lag polynomial $A(L)$ and imposing the no-bubbles condition yields a unique stationary solution characterized by the stable root $0 < \mu < 1$ of

$$\mathbb{P}(\mu) \equiv \beta \mu^2 - \left(1 + \beta + \frac{\kappa^2 \Xi}{\lambda_x} \right) \mu + 1 = 0.$$

Accordingly, the solution for $\Phi_{\pi_H,t}$ can be written recursively as

$$\Phi_{\pi_H,t} = \mu \Phi_{\pi_H,t-1} - \mu [(\sigma_u - \Gamma_{\pi_H,u})\varepsilon_t^u - \rho_u(u_t - u_{t-1})]. \quad (\text{A.9})$$

Using the AR(1) process for the cost-push shock,

$$u_t = \rho_u u_{t-1} + \sigma_u \varepsilon_t^u,$$

we obtain the moving-average representation

$$u_{t-1} = \sigma_u \sum_{j=0}^{\infty} \rho_u^j \varepsilon_{t-j-1}^u,$$

which implies

$$u_t - u_{t-1} = \sigma_u \varepsilon_t^u - (1 - \rho_u) \sigma_u \sum_{j=0}^{\infty} \rho_u^j \varepsilon_{t-j-1}^u.$$

Finally, setting $\rho_{\pi_H} \equiv \mu$ and using the commitment relation

$$\pi_{H,t} = \Phi_{\pi_H,t-1} + \Gamma_{\pi_H,u} \varepsilon_t^u$$

together with (A.9), we obtain

$$\pi_{H,t} = \rho_{\pi_H} \pi_{H,t-1} + \Gamma_{\pi_H,u} \varepsilon_t^u - \rho_{\pi_H} (1 - \rho_u) \sigma_u \sum_{j=0}^{\infty} \rho_u^j \varepsilon_{t-j-1}^u,$$

which establishes (3.4). $\square$

B Linear State Space System for Estimation

This section summarizes the transition and measurement equations used in Bayesian estimation.

Transition equation. Define the estimation state as

$$\begin{bmatrix} \xi_t \\ \xi_{t-1} \end{bmatrix} \equiv \begin{bmatrix} \varepsilon_t \\ \mathbf{X}_t \\ \Phi_{t-1} \\ \varepsilon_{t-1} \\ \mathbf{X}_{t-1} \\ \Phi_{t-2} \end{bmatrix}.$$

Using the solution matrices in (2.22), the transition equation is

$$\begin{bmatrix} \xi_{t+1} \\ \xi_t \end{bmatrix} = \tilde{H} \begin{bmatrix} \xi_t \\ \xi_{t-1} \end{bmatrix} + \nu_{t+1},$$

where

$$\tilde{H} \equiv \begin{bmatrix} H & 0 \\ I & 0 \end{bmatrix}, \quad \nu_{t+1} \equiv \begin{bmatrix} I \\ C_X \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \varepsilon_{t+1}.$$

Measurement equation. We estimate the model using two observables: CPI inflation π_t and nominal depreciation Δe_t . The measurement equation is

$$\begin{bmatrix} \pi_t \\ \Delta e_t \end{bmatrix} = \tilde{G} \begin{bmatrix} \xi_t \\ \xi_{t-1} \end{bmatrix} + \begin{bmatrix} \sigma_{\pi}^{me} & 0 \\ 0 & \sigma_{\Delta e}^{me} \end{bmatrix} \begin{bmatrix} \varepsilon_t^{me,\pi} \\ \varepsilon_t^{me,\Delta e} \end{bmatrix},$$

where $\tilde{G}$ in our model is given by

$$\tilde{G} = \begin{bmatrix} \Gamma_{\pi_H,u} + \alpha\tau\Gamma_{x,u} & \Gamma_{\pi_H,u} + \tau\Gamma_{x,u} \\ \Gamma_{\pi_H,a} + \alpha\tau\Gamma_{x,a} & \Gamma_{\pi_H,a} + \tau\Gamma_{x,a} \\ \Gamma_{\pi_H,y^*} + \alpha\tau\Gamma_{x,y^*} & \Gamma_{\pi_H,y^*} + \tau\Gamma_{x,y^*} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \alpha\tau\frac{1+\varphi}{\tau+\varphi} & \tau\frac{1+\varphi}{\tau+\varphi} \\ -\alpha\tau\frac{\tau+\varphi+(\omega-1)\alpha\tau}{\tau+\varphi} & -\tau\frac{\tau+\varphi+(\omega-1)\alpha\tau}{\tau+\varphi} \\ 1 & 1 \\ \alpha\tau & \tau \\ -\alpha\tau\Gamma_{x,u} & -\tau\Gamma_{x,u} \\ -\alpha\tau\Gamma_{x,a} & -\tau\Gamma_{x,a} \\ -\alpha\tau\Gamma_{x,y^*} & -\tau\Gamma_{x,y^*} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ -\alpha\tau\frac{1+\varphi}{\tau+\varphi} & -\tau\frac{1+\varphi}{\tau+\varphi} \\ \alpha\tau\frac{\tau+\varphi+(\omega-1)\alpha\tau}{\tau+\varphi} & \tau\frac{\tau+\varphi+(\omega-1)\alpha\tau}{\tau+\varphi} \\ 0 & 0 \\ -\alpha\tau & -\tau \end{bmatrix}^T,$$

where T denotes the transpose of the matrix.

C Closed-Economy Variant of the Model and Estimation

This appendix presents the closed-economy variant of the model underlying the analysis in Section 3.3, together with the corresponding linear state-space system used for its estimation. The Galí and Monacelli (2005) framework is isomorphic to a standard closed-economy New Keynesian model when home bias is complete ($\alpha = 0$). Under this restriction, the equilibrium conditions simplify as follows.

C.1 Equilibrium Conditions

Aggregate Demand. The slope parameter reduces to $\tau = \sigma$, and the aggregate demand equation becomes:

$$x_t = \hat{\mathbb{E}}_t x_{t+1} - \frac{1}{\sigma} \left(i_t - \hat{\mathbb{E}}_t \pi_{t+1} - r\bar{r}_t \right), \quad (\text{C.1})$$

where $\pi_t \equiv \pi_{H,t}$; that is, CPI inflation coincides with domestic inflation.

Natural Real Rate. The natural real rate depends only on productivity:

$$\bar{r}r_t = \rho + \Lambda_{r,a}^{closed} a_t, \quad (\text{C.2})$$

where $\Lambda_{r,a}^{closed} = -\sigma\Lambda_{y,a}^{closed}(1 - \rho_a)$, with $\Lambda_{y,a}^{closed}$ defined below.

Aggregate Supply. The hybrid New Keynesian aggregate supply retains the same functional form as in the open-economy case:

$$\pi_t = \tilde{\beta} \hat{\mathbb{E}}_t \pi_{t+1} + \eta \pi_{t-1} + \kappa x_t + u_t, \quad (\text{C.3})$$

where $\tilde{\beta}, \eta, \kappa$ follow the open-economy expressions for the hybrid weights, except that κ_0 in (2.12) is now specialized to its closed-economy form:

$$\kappa_0 \equiv \frac{(1 - \beta\zeta)(1 - \zeta)(\sigma + \varphi)}{\zeta}, \quad (\text{C.4})$$

i.e., τ in the open-economy expression for κ_0 is replaced by σ .

Natural Output. The natural level of output is determined by productivity alone:

$$\tilde{y}_t = \Lambda_{y,0} + \Lambda_{y,a}^{closed} a_t, \quad (\text{C.5})$$

where

$$\Lambda_{y,a}^{closed} = \frac{1 + \varphi}{\sigma + \varphi}. \quad (\text{C.6})$$

Shocks. The closed economy retains only *two* structural shocks: cost-push u_t and productivity a_t . The world-output shock y_t^* is absent. The two shocks follow AR(1) processes:

$$u_{t+1} = \rho_u u_t + \sigma_u \varepsilon_{t+1}^u, \quad (\text{C.7})$$

$$a_{t+1} = \rho_a a_t + \sigma_a \varepsilon_{t+1}^a, \quad (\text{C.8})$$

with $\varepsilon_t^u, \varepsilon_t^a$ i.i.d. standard normal innovations. The terms of trade s_t and the nominal exchange rate Δe_t are absent from the equilibrium system.

Robustly Optimal Monetary Policy. The belief-distortion framework and the central bank's robustly optimal monetary policy problem in Section 2.4 carry over without modification, with the relevant system matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$ replaced by their closed-economy counterparts (i.e., with the y_t^*

row/column block removed). In particular, the central bank solves the following min–max problem,

$$\min_{\{\pi_t, x_t, i_t\}} \max_{\{m_{t+1}\}} \mathbb{E}_{-1} \sum_{t=0}^{\infty} \beta^t \frac{1}{2} (\pi_t^2 + \lambda_x (x_t - \bar{x})^2) - \theta \mathbb{E}_{-1} \sum_{t=0}^{\infty} \beta^t m_{t+1} \ln m_{t+1},$$

subject to (C.1), (C.2), (C.3), (C.7), (C.8), and $\mathbb{E}_t m_{t+1} = 1$.

C.2 Estimation

We re-estimate both the NRE and RE models under the closed-economy restriction. The estimation results are reported and discussed in Section 3.3; the priors for the parameters common to the open- and closed-economy specifications are identical to those of the baseline estimation in Table E.10.

Data. We use CPI inflation, π_t —the same observable as in the open-economy estimation—and quarterly output growth, Δy_t , as observables. Output growth is defined as $\Delta y_t \equiv \log Y_t - \log Y_{t-1}$, where Y_t is seasonally adjusted real GDP per capita for Mexico over the sample period 1Q 1998–4Q 2007.¹ In the closed-economy model, CPI inflation is directly identified with the model’s inflation variable, $\pi_t = \pi_{H,t}$. The output growth observable relates to model variables through the identity

$$\Delta y_t = (x_t - x_{t-1}) + \Lambda_{y,a}^{closed} (a_t - a_{t-1}), \quad (\text{C.9})$$

i.e., the sum of changes in the output gap and changes in natural output (driven by TFP). Given the observables $\{\pi_t, \Delta y_t\}$, the measurement identity (C.9) and the aggregate supply relation (C.3) discipline the joint dynamics of inflation, the output gap, and TFP, while the remaining equilibrium conditions pin down the implied policy rate.

The Linear State-Space System for Estimation. Define

$$\begin{aligned} \varepsilon_t &= [\varepsilon_t^u, \varepsilon_t^a]', \\ \mathbf{X}_t &= [1, \pi_{t-1}, u_t, a_t]', \\ \mathbf{Y}_t &= [\pi_t, x_t]', \end{aligned}$$

and

$$\boldsymbol{\xi}_t \equiv [\varepsilon_t', \mathbf{X}_t', \boldsymbol{\Phi}_{t-1}']'.$$

Note that, relative to the open-economy case, $\mathbf{X}_t$ omits the world-output state y_t^* , and ε_t omits the corresponding innovation $\varepsilon_t^{y^*}$. The forward-looking block $\mathbf{Y}_t = [\pi_t, x_t]'$ retains the same dimension.

¹We construct real GDP per capita by obtaining seasonally adjusted real gross domestic product for Mexico (FRED series NGDPRSA XDCMXQ) and then dividing it by total population (FRED series POPTOTMXA647NWDB; annual, held constant within each year).

Because the measurement equation for Δy_t involves lagged model variables (in particular a_{t-1} and x_{t-1}), we use $[\xi'_t, \xi'_{t-1}]'$ as the state vector for the measurement system. The resulting measurement equation, allowing for measurement errors, is:

$$\begin{bmatrix} \pi_t \\ \Delta y_t \end{bmatrix} = \tilde{\mathbf{G}}^{closed} \begin{bmatrix} \xi_t \\ \xi_{t-1} \end{bmatrix} + \begin{bmatrix} \sigma_\pi^{me} & 0 \\ 0 & \sigma_{\Delta y}^{me} \end{bmatrix} \begin{bmatrix} \epsilon_t^{me,\pi} \\ \epsilon_t^{me,\Delta y} \end{bmatrix}, \quad (\text{C.10})$$

where $[\epsilon_t^{me,\pi}, \epsilon_t^{me,\Delta y}]'$ are orthogonal standard normal random variables, and $[\sigma_\pi^{me}, \sigma_{\Delta y}^{me}]'$ are the standard deviations of the measurement errors. The corresponding transition equation is:

$$\begin{bmatrix} \xi_{t+1} \\ \xi_t \end{bmatrix} = \tilde{\mathbf{H}}^{closed} \begin{bmatrix} \xi_t \\ \xi_{t-1} \end{bmatrix} + \nu_{t+1}, \quad (\text{C.11})$$

which is structurally analogous to the open-economy transition equation (3.3), but with the dimensions and matrices adjusted to remove the y_t^* block. Equations (C.10) and (C.11) together define the linear state-space system used for the closed-economy estimation.

The State-Space Matrices. Under the self-consistent conditionally linear commitment, the forward-looking block satisfies

$$\mathbf{Y}_t = \Phi_{t-1} + \mathbf{\Gamma} \varepsilon_t, \quad \text{where} \quad \mathbf{\Gamma} \equiv \begin{bmatrix} \Gamma_{\pi,u} & \Gamma_{\pi,a} \\ \Gamma_{x,u} & \Gamma_{x,a} \end{bmatrix}. \quad (\text{C.12})$$

The measurement equation then follows from the two observation identities: $\pi_t = \Phi_{\pi,t-1} + \Gamma_{\pi,u} \varepsilon_t^u + \Gamma_{\pi,a} \varepsilon_t^a$, and

$$\Delta y_t = (x_t - x_{t-1}) + \Lambda_{y,a}^{closed} (a_t - a_{t-1}),$$

with x_t and x_{t-1} substituted from (C.12). All entries of the resulting measurement matrix $\tilde{\mathbf{G}}^{closed}$ depend only on $\mathbf{\Gamma}$ and $\Lambda_{y,a}^{closed}$. The transition equation takes the same form as in Appendix B: the law of motion

$$\xi_{t+1} = \mathbf{H}^{closed} \xi_t + \mathbf{S}^{\varepsilon,closed} \varepsilon_{t+1}, \quad \mathbf{S}^{\varepsilon,closed} = \begin{bmatrix} \mathbf{I} \\ \mathbf{C}_X^{closed} \\ \mathbf{0} \end{bmatrix}, \quad (\text{C.13})$$

implies

$$\tilde{\mathbf{H}}^{closed} = \begin{bmatrix} \mathbf{H}^{closed} & \mathbf{0} \\ \mathbf{I} & \mathbf{0} \end{bmatrix}, \quad \nu_{t+1} = \begin{bmatrix} \mathbf{S}^{\varepsilon,closed} \\ \mathbf{0} \end{bmatrix} \varepsilon_{t+1},$$

where $\mathbf{C}_X^{closed}$ is the $\mathbf{X}$ -block of the closed-economy $\mathbf{C}$ matrix. Given posterior parameter draws $\Theta^{closed} = [\tilde{\theta}, \delta, \varphi, \rho_u, \sigma_u, \rho_a, \sigma_a]'$, the matrices $\mathbf{H}^{closed}$, $\mathbf{G}^{closed}$, and $\mathbf{\Gamma}$ are computed numerically as the solution of the closed-economy Kwon and Miao (2019) fixed-point iteration—through which $\tilde{\theta}$

enters the evolution of Φ_t —and the Kalman filter is then run on (C.10)–(C.11).

Identification of $\tilde{\theta}$ in the Closed Economy. With CPI inflation and output growth as observables, the central bank’s concern for robustness, $\tilde{\theta}$, is identified through the joint time-series dynamics of inflation and output growth. The nominal depreciation rate is replaced by the output growth rate. Thus, the closed-economy version uses different observables and does not rely on the open-economy restrictions such as the risk-sharing condition (which relates the output gap to the terms of trade) and the depreciation identity.

D The PRE Model and Estimation

This appendix derives the state-space representation underlying the PRE model discussed in Section 4.1, in which the central bank commits to the robustly optimal reaction function derived under worst-case beliefs, while private agents actually form expectations rationally. It then casts this law of motion into the state-space form on which the estimation is based.

D.1 The Stacked Linear System

Central Bank. In the PRE model, the central bank continues to set the nominal interest rate according to the same robustly optimal reaction function assuming that private agents hold the worst-case beliefs. Concretely, let $\mathbf{G}_{1,\bullet}$ denote the row of $\mathbf{G}$ associated with the interest rate, the policy instrument, in Equation (2.23). Then, the central bank commits to

$$i_t^{CB} = \mathbf{G}_{1,\bullet} \boldsymbol{\xi}_t, \quad (\text{D.1})$$

where $\boldsymbol{\xi}_t \equiv [\boldsymbol{\varepsilon}'_t, \mathbf{X}'_t, \Phi'_{t-1}]'$ is exactly the state vector of Section 2.4, and the coefficients of $\mathbf{G}_{1,\bullet}$ are also unchanged.

Next, the stochastic intercept of the conditionally linear commitment rule, Φ_t , is also kept the same. Since $\mathbf{Y}_{t+1} = \Phi_t + \mathbf{\Gamma} \boldsymbol{\varepsilon}_{t+1}$, the intercept Φ_t carries a precise dual interpretation once the robustly optimal policy has been solved. Because $\boldsymbol{\varepsilon}_{t+1}$ has mean zero under the approximating measure $\mathcal{P}$, taking expectations of the commitment rule gives $\mathbb{E}_t \mathbf{Y}_{t+1} = \Phi_t$: Φ_t is the central bank’s own, undistorted forecast of next period’s inflation and output gap. At the same time, equation (2.25) shows that $\Phi_t + \tilde{\theta} \mathbf{\Gamma} \mathbf{\Gamma}' \boldsymbol{\mu}_{Y,t} = \mathbb{E}_t m_{t+1} \mathbf{Y}_{t+1} = \hat{\mathbb{E}}_t \mathbf{Y}_{t+1}$: the same Φ_t , shifted by the entropy-weighted tilt $\tilde{\theta} \mathbf{\Gamma} \mathbf{\Gamma}' \boldsymbol{\mu}_{Y,t}$, equals the worst-case forecast that private agents actually use when their beliefs coincide with the central bank’s worst case, as in the baseline NRE model. However, Φ_t is not a primitive of the model: it is computed once from the central bank’s first-order conditions (Appendix A.2), and its law of motion,

$$\Phi_t = \mathbf{H}_{\Phi,\bullet} \boldsymbol{\xi}_t \quad (\text{D.2})$$

is a byproduct of that same [Klein \(2000\)](#) solution. It makes no reference to which beliefs private agents actually hold, only to the central bank's own optimization problem. This is precisely what makes Φ_t portable to the PRE model.

In summary, under PRE the central bank does not re-solve its optimization problem: it has already committed, under the presumption of worst-case beliefs. Thus, the reaction function and the recursion of the co-state variable both are carried over unchanged, coefficient for coefficient, from the NRE solution as depicted in Equations (D.1) and (D.2).

Private Agents. The private agents' equilibrium conditions in the PRE model are simply the rational-expectations counterparts of (2.7) and (2.10), with $\hat{\mathbb{E}}_t[\cdot]$ replaced by $\mathbb{E}_t[\cdot]$. The structural matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$ and their partitions $\mathbf{A}_{XX}, \mathbf{A}_{XY}, \mathbf{A}_{YX}, \mathbf{A}_{YY}, \mathbf{B}_X, \mathbf{B}_Y, \mathbf{C}_X, \mathbf{C}_Y$ are identical to those reported in Appendix A.2. What differs from Section 2.4 is that i_t is no longer a free choice variable: the central bank has already committed to rule (D.1), which we partition conformably with ξ_t as $i_t = \mathbf{G}_{1,\varepsilon}\varepsilon_t + \mathbf{G}_{1,X}\mathbf{X}_t + \mathbf{G}_{1,\Phi}\Phi_{t-1}$. Likewise, the co-state recursion (D.2) partitions as $\Phi_t = \mathbf{H}_{\Phi,\varepsilon}\varepsilon_t + \mathbf{H}_{\Phi,X}\mathbf{X}_t + \mathbf{H}_{\Phi,\Phi}\Phi_{t-1}$, with $\mathbf{H}_{\Phi,\varepsilon}, \mathbf{H}_{\Phi,X}, \mathbf{H}_{\Phi,\Phi}$ the corresponding partitions of $\mathbf{H}$ from (2.22). Substituting the partitioned rule into the definitions of $\mathbf{X}_{t+1}$ and $\mathbb{E}_t\mathbf{Y}_{t+1}$, and appending the co-state recursion to keep track of the central bank's commitment state, yields the stacked system:

$$\begin{bmatrix} \mathbb{E}_t\varepsilon_{t+1} \\ \mathbb{E}_t\mathbf{X}_{t+1} \\ \Phi_t \\ \mathbb{E}_t\mathbf{Y}_{t+1} \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_X\mathbf{G}_{1,\varepsilon} & \mathbf{A}_{XX} + \mathbf{B}_X\mathbf{G}_{1,X} & \mathbf{B}_X\mathbf{G}_{1,\Phi} & \mathbf{A}_{XY} \\ \mathbf{H}_{\Phi,\varepsilon} & \mathbf{H}_{\Phi,X} & \mathbf{H}_{\Phi,\Phi} & \mathbf{0} \\ \mathbf{B}_Y\mathbf{G}_{1,\varepsilon} & \mathbf{A}_{YX} + \mathbf{B}_Y\mathbf{G}_{1,X} & \mathbf{B}_Y\mathbf{G}_{1,\Phi} & \mathbf{A}_{YY} \end{bmatrix} \begin{bmatrix} \varepsilon_t \\ \mathbf{X}_t \\ \Phi_{t-1} \\ \mathbf{Y}_t \end{bmatrix}, \quad (\text{D.3})$$

where $\{\varepsilon_t, \mathbf{X}_t, \Phi_{t-1}\}$ are predetermined and $\mathbf{Y}_t = [\pi_{H,t}, x_t]'$ is the non-predetermined block, now pinned down entirely by the rational-expectations equilibrium conditions rather than by central-bank optimization. System (D.3) has the standard form of a linear rational-expectations system in the sense of [Klein \(2000\)](#), since the interest-rate rule and the co-state law of motion are both known functions of the predetermined state.

D.2 Solution and State-Space Representation

Solving (D.3) delivers the actual law of motion of the economy under PRE,²

$$\begin{bmatrix} \varepsilon_{t+1} \\ \mathbf{X}_{t+1} \\ \Phi_t \end{bmatrix} = \mathbf{H}^{actual} \begin{bmatrix} \varepsilon_t \\ \mathbf{X}_t \\ \Phi_{t-1} \end{bmatrix} + \mathbf{S}^{\varepsilon, actual} \varepsilon_{t+1}, \quad \mathbf{Y}_t = \mathbf{G}^{actual} \begin{bmatrix} \varepsilon_t \\ \mathbf{X}_t \\ \Phi_{t-1} \end{bmatrix}, \quad (\text{D.4})$$

with $\mathbf{S}^{\varepsilon, actual} = \mathbf{S}^\varepsilon \equiv [\mathbf{I}, \mathbf{C}'_X, \mathbf{0}']'$, unchanged from (2.22). Equation (D.4) replaces (2.22)–(2.23) as the law of motion that actually governs the economy under PRE; and it generally differs from the law of motion $\xi_{t+1} = \mathbf{H}\xi_t + \mathbf{S}^\varepsilon \varepsilon_{t+1}$, $\mathbf{Y}_t = \tilde{\mathbf{G}}\xi_t$, that the central bank uses to design and evaluate its policy, since the latter is derived under the presumption of worst-case beliefs. This mismatch has a direct and interpretable implication: it opens a wedge between the private sector's actual forecast of $\mathbf{Y}_{t+1}$ and the forecast that the central bank's promise-keeping variable Φ_t is designed to track:

$$\mathbb{E}_t \mathbf{Y}_{t+1} - \Phi_t = \left(\mathbf{G}^{actual} \mathbf{H}^{actual} - \mathbf{H}_{\Phi, \bullet} \right) \xi_t.$$

This gap quantifies, period by period, how far the rational-expectations equilibrium departs from the worst-case scenario the central bank guards against. Under the baseline NRE model the gap is zero by construction, since Φ_t is precisely $\mathbb{E}_t \mathbf{Y}_{t+1}$. Additionally, in the worst-case scenario, $\mathbf{G}^{actual} \mathbf{H}^{actual} \xi_t = [\Gamma, \mathbf{0}, \mathbf{I}] \mathbf{H} \xi_t$ which is likewise equivalent to $\mathbb{E}_t \mathbf{Y}_{t+1}$. Under PRE, however, the gap need not vanish, and its magnitude offers a concrete, interpretable metric for the potential cost of a misplaced concern for robustness.³

D.3 Measurement System for Estimation

Estimation of the PRE model follows Appendix B exactly, with $\xi_t \equiv [\varepsilon'_t, \mathbf{X}'_t, \Phi'_{t-1}]'$ as before. Since the transition equation utilizes $[\xi'_t, \xi'_{t-1}]'$ as the estimation state vector, the corresponding matrix $\tilde{\mathbf{H}}^{actual}$ will be given as $[\mathbf{H}^{actual}, \mathbf{0}; \mathbf{I}, \mathbf{0}]$. Next, the measurement equations for π_t and Δe_t are constructed from $\mathbf{Y}_t$ and its lag exactly as in (3.1) and (2.15), the only change relative to the baseline model is that $\mathbf{Y}_t$ is now given by $\mathbf{G}^{actual} \xi_t$ in (D.4) rather than by $[\Gamma, \mathbf{0}, \mathbf{I}] \xi_t$. The Kalman filter and the random-walk Metropolis-Hastings algorithm are then applied exactly as described in Section 3, using the same priors (Table E.10) and the same observables $(\pi_t, \Delta e_t)$ over 1Q 1998–4Q 2007.

²With two jump variables and a committed policy rule that responds only to predetermined states, the Blanchard–Kahn root count does not deliver a unique bounded solution: the system has one more stable direction than predetermined variables, so more than one bounded solution satisfies the equilibrium conditions. This feature is driven by the absence of feedback from the current jump variables in the committed rule, the same force that generates indeterminacy under interest-rate rules that do not respond to current inflation. We therefore solve directly for the coefficients of $\mathbf{Y}_t$, seeding the solution at $\mathbf{Y}_t = \Phi_{t-1} + \Gamma \varepsilon_t$, the exact solution at $\tilde{\theta} = 0$, where the PRE model collapses to the RE model under the optimal commitment rule. This procedure selects the unique branch that varies continuously with $\tilde{\theta}$ and coincides with the known solution at $\tilde{\theta} = 0$; details of the solution procedure are provided in the replication files.

³We thank an anonymous referee for raising this point.

E Additional Tables and Figures

Table E.8: Parameters

Parameter	Description	Remarks
β	Subjective discount factor	0.975 (MEX), 0.99 (CAN)
$\tilde{\theta}$	Central bank's degree of concern for robustness with respect to private agents' NRE	Estimated (Country- and model-specific)
ρ	Steady-state real interest rate $\beta^{-1} - 1$	0.0256 (MEX), 0.0101 (CAN)
σ	Curvature of the utility function	1
α	Degree of home bias (openness of the economy)	0.279 (MEX), 0.33 (CAN)
ϵ	Elasticity of substitution across home varieties	6
ζ	Calvo–Yun price stickiness parameter	0.75
φ	Inverse Frisch elasticity of labor supply	Estimated (Country- and model-specific)
ω	Effect of terms-of-trade movements on output	Estimated (Country- and model-specific)
Θ	Composite openness parameter $\omega - 1$	Implied (Country- and model-specific)
τ	Slope of the aggregate demand curve $\frac{\sigma}{1+\alpha(\omega-1)}$	Implied (Country- and model-specific)
κ_0	Baseline slope of the aggregate supply curve $\frac{(1-\beta\zeta)(1-\zeta)(\tau+\varphi)}{\zeta}$	Implied (Country- and model-specific)
δ	Fraction of backward-looking firms	Estimated (Country- and model-specific)
κ	Slope of the hybrid Phillips curve $\frac{\zeta(1-\delta)\kappa_0}{[\zeta+\delta(1-\zeta(1-\beta))]}$	Implied (Country- and model-specific)
$\tilde{\beta}$	Effective discount factor in the hybrid Phillips curve $\frac{\zeta\beta}{[\zeta+\delta(1-\zeta(1-\beta))]}$	Implied (Country- and model-specific)
η	Weight on lagged inflation in the hybrid Phillips curve $\frac{\delta}{[\zeta+\delta(1-\zeta(1-\beta))]}$	Implied (Country- and model-specific)
λ_x	Relative weight on output gap stabilization in the loss function $\frac{(1-\beta\zeta)(1-\zeta)(\tau+\varphi)}{\zeta\epsilon}$	Implied (Country- and model-specific)
$\bar{x}$	Long-run output gap target	0
$\Lambda_{y,0}$	Intercept on the natural level of output	0
$\Lambda_{y,a}$	Effect of TFP on the natural level of output $\frac{1+\varphi}{\tau+\varphi}$	Implied (Country- and model-specific)
Λ_{y,y^*}	Effect of world output on the natural level of output $-\frac{\alpha\Theta\tau}{\tau+\varphi}$	Implied (Country- and model-specific)
$\Lambda_{r,a}$	Effect of TFP on the natural interest rate $-\tau\Lambda_{y,a}(1-\rho_a)$	Implied (Country- and model-specific)
Λ_{r,y^*}	Effect of world output on the natural interest rate $-\varphi\Lambda_{y,y^*}(1-\rho_{y^*})$	Implied (Country- and model-specific)
ρ_u	Persistence of cost-push shock	Estimated (Country- and model-specific)
ρ_a	Persistence of TFP shock	Estimated (Country- and model-specific)
ρ_{y^*}	Persistence of world output shock	0.84 (MEX), 0.85 (CAN)
σ_u	Standard deviation of cost-push shock	Estimated (Country- and model-specific)
σ_a	Standard deviation of TFP shock	Estimated (Country- and model-specific)
σ_{y^*}	Standard deviation of world output shock	0.0045 (MEX), 0.0042 (CAN)
σ_{π}^{me}	Standard deviation of measurement error in CPI inflation	Estimated (Country- and model-specific)
$\sigma_{\Delta e}^{me}$	Standard deviation of measurement error in nominal depreciation	Estimated (Country- and model-specific)

Note. “Implied” indicates that the parameter is determined as a function of calibrated and estimated parameters. “Country- and model-specific” indicates that the parameter values vary across models (NRE and RE) and countries (Mexico and Canada).

Table E.9: Pre-Set Parameters: Mexico

σ	1
ζ	0.75
ρ_{y^*}	0.84
$\sigma_{y^*} \times 100$	0.45
β	0.975
α	0.279

Note. The time unit is one quarter.

Table E.10: Prior Distributions of Parameters for Estimation

Parameter	Support	Distribution	Para (1)	Para (2)	[mean, std]
$\tilde{\theta}$	$[0, 10^4]$	Uniform	0	10^4	[5000, 2886.75]
δ	$[0, 1)$	Beta	2.0	3.0	[0.4, 0.2]
ω	$\mathbb{R}^+$	Gamma	2.0	2.0	[4.0, 2.82]
φ	$\mathbb{R}^+$	Gamma	2.0	2.0	[4.0, 2.82]
ρ_u	$[0, 1)$	Beta	2.0	3.0	[0.4, 0.2]
ρ_a	$[0, 1)$	Beta	2.0	3.0	[0.4, 0.2]
$\sigma_u \times 100$	$\mathbb{R}^+$	Gamma	0.25	2.0	[0.5, 1.0]
$\sigma_a \times 100$	$\mathbb{R}^+$	Gamma	0.25	2.0	[0.5, 1.0]

Table E.11: Posterior Distributions: Closed-Economy Estimation

Parameters	NRE		RE		NRE ₀		RE ₀	
	Mean	[5%, 95%]	Mean	[5%, 95%]	Mean	[5%, 95%]	Mean	[5%, 95%]
$\tilde{\theta}$	0.076	[0.004, 0.233]	-	-	34.33	[2.42, 98.30]	-	-
δ	0.96	[0.94, 0.98]	0.97	[0.95, 0.99]	-	-	-	-
φ	12.90	[6.52, 21.16]	9.04	[4.97, 13.26]	0.38	[0.05, 0.97]	0.38	[0.06, 0.97]
ρ_u	0.27	[0.09, 0.48]	0.23	[0.07, 0.42]	0.94	[0.90, 0.97]	0.95	[0.91, 0.98]
σ_u	0.0020	[0.0013, 0.0028]	0.0020	[0.0013, 0.0027]	0.0097	[0.0070, 0.0139]	0.0096	[0.0069, 0.0137]
ρ_a	0.92	[0.86, 0.97]	0.91	[0.85, 0.97]	0.95	[0.92, 0.98]	0.95	[0.92, 0.98]
σ_a	0.0104	[0.0083, 0.0128]	0.0101	[0.0080, 0.0125]	0.0654	[0.0533, 0.0795]	0.0636	[0.0521, 0.0771]
Log MDD	253.21		265.37		160.68		166.79	

Note. The labels follow Table 1: ‘NRE₀’ and ‘RE₀’ refer to the NRE and RE models, which are estimated by imposing no backward-looking firms, i.e., $\delta = 0$. The closed-economy specifications impose $\alpha = 0$ and use CPI inflation and output growth as observables (see Appendix C); priors are identical to the baseline estimation. The open-economy parameter ω is absent from the closed-economy model, and standard deviations of measurement errors are omitted from the table. The posterior mean, 5th percentile, and 95th percentile were computed over one million draws from the MCMC chain.

Table E.12: Posterior Distributions: PRE Estimation

Parameters	PRE		PRE ₀	
	Mean	[5%, 95%]	Mean	[5%, 95%]
$\tilde{\theta}$	341.76	[189.10, 560.90]	327.09	[137.12, 581.43]
δ	0.10	[0.02, 0.20]	-	-
ω	9.00	[6.71, 12.1]	5.80	[3.75, 8.91]
φ	4.62	[2.24, 8.70]	1.33	[0.13, 4.61]
ρ_u	0.69	[0.57, 0.79]	0.44	[0.18, 0.75]
σ_u	0.055	[0.035, 0.083]	0.028	[0.016, 0.062]
ρ_a	0.89	[0.77, 0.97]	0.88	[0.75, 0.96]
σ_a	0.035	[0.023, 0.051]	0.021	[0.011, 0.037]
Log MDD	208.19		208.29	

Note. The label ‘PRE’ refers to the model in Section 4.1, which retains the central bank’s robust rule but endows private agents with rational expectations. The label ‘PRE₀’ refers to the PRE model, which is estimated by imposing no backward-looking firms, i.e., $\delta = 0$. The observables and priors are identical to the baseline estimation. Standard deviations of measurement errors are omitted from the table. The posterior mean, 5th percentile, and 95th percentile were computed over one million draws from the MCMC chain.

Table E.13: Posterior Distributions under Alternative Priors for $\tilde{\theta}$

Parameters	Uniform $[0, 10^6]$				Uniform $[0, 10^{18}]$			
	NRE		NRE ₀		NRE		NRE ₀	
	Mean	[5%, 95%]	Mean	[5%, 95%]	Mean	[5%, 95%]	Mean	[5%, 95%]
$\tilde{\theta}$	1801	[905, 3113]	1591	[825, 2685]	1789	[898, 3101]	1588	[822, 2686]
δ	0.10	[0.02, 0.22]	–	–	0.10	[0.02, 0.21]	–	–
ω	8.18	[6.17, 11.0]	8.13	[6.00, 11.2]	8.17	[6.14, 11.0]	8.07	[5.98, 11.1]
φ	1.96	[0.48, 4.88]	1.29	[0.30, 3.15]	1.87	[0.47, 4.52]	1.29	[0.30, 3.19]
ρ_u	0.32	[0.09, 0.58]	0.38	[0.15, 0.62]	0.32	[0.09, 0.57]	0.37	[0.15, 0.62]
σ_u	0.029	[0.016, 0.055]	0.028	[0.016, 0.052]	0.029	[0.016, 0.053]	0.028	[0.016, 0.052]
ρ_a	0.88	[0.77, 0.96]	0.88	[0.77, 0.96]	0.88	[0.77, 0.96]	0.88	[0.77, 0.96]
σ_a	0.031	[0.018, 0.048]	0.027	[0.015, 0.044]	0.031	[0.017, 0.049]	0.027	[0.015, 0.043]

Note. The labels follow Table 1: ‘NRE₀’ refers to the NRE model, which is estimated by imposing no backward-looking firms, i.e., $\delta = 0$. The two column blocks replace the baseline prior for $\tilde{\theta}$ with the indicated uniform priors; all other priors, the observables, and the estimation procedure are identical to the baseline estimation. Standard deviations of measurement errors are omitted from the table. The posterior mean, 5th percentile, and 95th percentile were computed over one million draws from the MCMC chain.

Table E.14: Posterior Distributions by Extended Sample Period

(a) 1Q 1996–4Q 2007

Parameters	NRE		RE		NRE ₀		RE ₀	
	Mean	[5%, 95%]	Mean	[5%, 95%]	Mean	[5%, 95%]	Mean	[5%, 95%]
$\tilde{\theta}$	1035	[544, 1885]	–	–	959	[508, 1728]	–	–
δ	0.06	[0.01, 0.16]	0.54	[0.21, 0.87]	–	–	–	–
ω	13.7	[9.81, 18.9]	7.82	[2.12, 11.0]	13.3	[9.52, 18.4]	8.92	[7.18, 11.4]
φ	1.80	[0.68, 4.00]	0.56	[0.07, 4.00]	1.31	[0.49, 2.76]	0.14	[0.03, 0.38]
ρ_u	0.60	[0.43, 0.73]	0.86	[0.19, 0.95]	0.62	[0.45, 0.74]	0.94	[0.88, 0.97]
σ_u	0.037	[0.019, 0.069]	0.006	[0.003, 0.010]	0.033	[0.018, 0.062]	0.006	[0.004, 0.009]
ρ_a	0.96	[0.92, 0.98]	0.87	[0.72, 0.95]	0.96	[0.92, 0.98]	0.81	[0.68, 0.92]
σ_a	0.054	[0.034, 0.081]	0.040	[0.028, 0.065]	0.048	[0.030, 0.073]	0.035	[0.027, 0.048]

(b) 1Q 1997–4Q 2007

Parameters	NRE		RE		NRE ₀		RE ₀	
	Mean	[5%, 95%]	Mean	[5%, 95%]	Mean	[5%, 95%]	Mean	[5%, 95%]
$\tilde{\theta}$	1868	[961, 3633]	–	–	1685	[875, 3293]	–	–
δ	0.08	[0.02, 0.21]	0.73	[0.43, 0.87]	–	–	–	–
ω	11.3	[8.17, 15.7]	2.84	[0.80, 7.40]	10.8	[7.87, 15.1]	8.21	[6.02, 11.4]
φ	2.29	[0.83, 5.55]	2.06	[0.29, 7.16]	1.41	[0.50, 3.34]	0.26	[0.05, 0.75]
ρ_u	0.48	[0.26, 0.65]	0.38	[0.07, 0.88]	0.49	[0.28, 0.66]	0.91	[0.83, 0.96]
σ_u	0.035	[0.019, 0.067]	0.009	[0.006, 0.014]	0.030	[0.017, 0.058]	0.007	[0.005, 0.012]
ρ_a	0.93	[0.84, 0.97]	0.90	[0.81, 0.96]	0.93	[0.85, 0.98]	0.84	[0.70, 0.94]
σ_a	0.044	[0.028, 0.066]	0.034	[0.024, 0.054]	0.039	[0.024, 0.059]	0.038	[0.028, 0.057]

Note. The labels follow Table 1: ‘NRE₀’ and ‘RE₀’ refer to the NRE and RE models, which are estimated by imposing no backward-looking firms, i.e., $\delta = 0$. Each panel re-estimates the four specifications over the indicated sample period; the observables, priors, and estimation procedure are identical to the baseline estimation. Standard deviations of measurement errors are omitted from the table. The posterior mean, 5th percentile, and 95th percentile were computed over one million draws from the MCMC chain.

Table E.15: Estimation Results: Pre- and Post-Inflation-Targeting Sub-Samples

Parameters	Pre-adoption (1Q 1998–4Q 2001)				Post-adoption (1Q 2002–4Q 2007)			
	NRE		RE		NRE		RE	
	Mean	[5%, 95%]	Mean	[5%, 95%]	Mean	[5%, 95%]	Mean	[5%, 95%]
θ	3970	[1738, 7373]	-	-	1098	[159, 2585]	-	-
δ	0.17	[0.04, 0.31]	0.51	[0.17, 0.77]	0.12	[0.03, 0.23]	0.07	[0.01, 0.15]
ω	7.49	[5.36, 10.21]	4.46	[1.61, 8.23]	2.86	[2.16, 3.69]	2.46	[1.92, 3.05]
φ	4.53	[1.85, 8.63]	3.27	[0.34, 8.42]	4.37	[1.06, 9.37]	3.53	[0.67, 8.01]
ρ_u	0.29	[0.10, 0.50]	0.40	[0.11, 0.73]	0.33	[0.09, 0.60]	0.48	[0.19, 0.75]
σ_u	0.042	[0.024, 0.066]	0.013	[0.007, 0.022]	0.028	[0.015, 0.048]	0.027	[0.015, 0.046]
ρ_a	0.51	[0.16, 0.82]	0.77	[0.57, 0.91]	0.77	[0.54, 0.92]	0.76	[0.53, 0.91]
σ_a	0.026	[0.003, 0.049]	0.055	[0.034, 0.084]	0.012	[0.008, 0.017]	0.011	[0.007, 0.015]
Log MDD	78.41		68.99		145.20		146.72	
Policy-rate RMSE	0.0443		0.1028		0.0529		0.0549	

Note. The labels ‘NRE’ and ‘RE’ refer to the baseline models in Section 2, estimated with backward-looking firms. The sample is split at the formal adoption of the inflation-targeting regime in 2001, and each sub-sample is estimated separately; priors and the estimation procedure are identical to the full-sample baseline. The policy-rate RMSE is computed for the demeaned, annualized policy rate, as in Figure 3. Standard deviations of measurement errors are omitted from the table. The posterior mean, 5th percentile, and 95th percentile were computed over one million draws from the MCMC chain.

Table E.16: Posterior Distributions: Canada

Parameters	NRE		RE		NRE ₀		RE ₀	
	Mean	[5%, 95%]	Mean	[5%, 95%]	Mean	[5%, 95%]	Mean	[5%, 95%]
θ	49.5	[14.4, 94.3]	-	-	37.4	[10.4, 70.4]	-	-
δ	0.06	[0.01, 0.12]	0.08	[0.016, 0.21]	-	-	-	-
ω	4.22	[3.38, 5.12]	3.35	[2.86, 3.81]	3.93	[3.13, 4.83]	3.45	[2.89, 3.98]
φ	2.55	[0.77, 5.23]	2.18	[0.51, 4.82]	1.65	[0.44, 3.44]	1.68	[0.45, 3.53]
ρ_u	0.92	[0.86, 0.96]	0.95	[0.91, 0.98]	0.91	[0.86, 0.96]	0.94	[0.89, 0.98]
σ_u	0.026	[0.014, 0.045]	0.020	[0.011, 0.034]	0.022	[0.012, 0.038]	0.021	[0.012, 0.036]
ρ_a	0.89	[0.80, 0.96]	0.93	[0.88, 0.97]	0.88	[0.79, 0.95]	0.92	[0.86, 0.97]
σ_a	0.013	[0.009, 0.017]	0.012	[0.008, 0.015]	0.012	[0.008, 0.016]	0.011	[0.008, 0.014]
Log MDD	416.23		419.34		420.17		423.15	

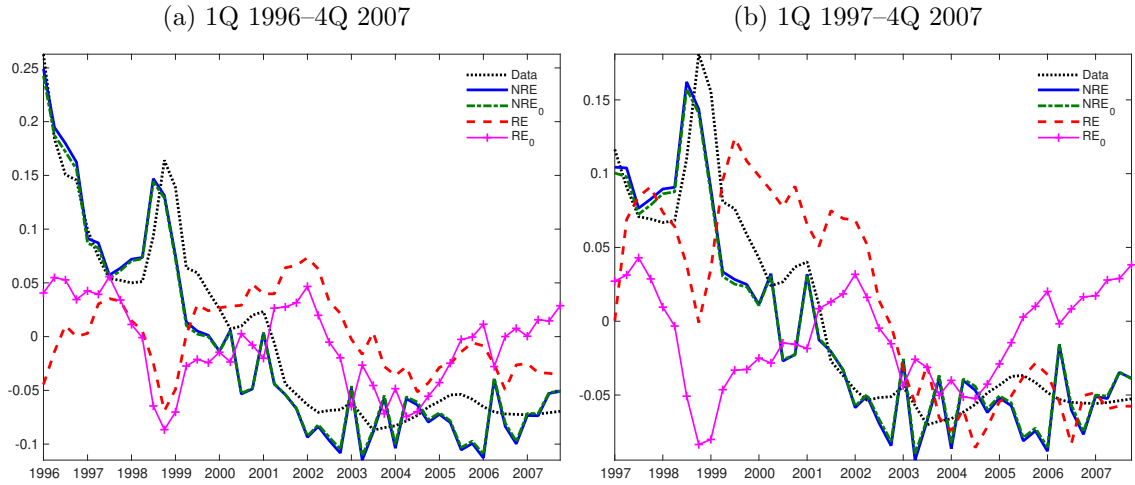
Note. The labels follow Table 1: ‘NRE₀’ and ‘RE₀’ refer to the NRE and RE models, which are estimated by imposing no backward-looking firms, i.e., $\delta = 0$. The models are estimated on Canadian data over the sample period 2Q 1991–4Q 2007, with observables and priors identical to the baseline estimation. Standard deviations of measurement errors are omitted from the table. The posterior mean, 5th percentile, and 95th percentile were computed over one million draws from the MCMC chain.

Table E.17: Estimation Results: Sensitivity to Pre-Set Parameters β and ζ

β	ζ	$\hat{\theta}$ (NRE)		$\hat{\theta}$ (NRE ₀)		δ		$\Delta\text{Log MDD}$		Policy-rate RMSE			
		Mean	[5%, 95%]	Mean	[5%, 95%]	NRE	RE	NRE-RE	NRE ₀ -RE ₀	NRE	RE	NRE ₀	RE ₀
0.975	0.75	1807	[896, 3190]	1594	[831, 2708]	0.10	0.49	17.5	20.8	0.0295	0.0788	0.0301	0.0910
0.99	0.75	2175	[1399, 3134]	1910	[880, 3269]	0.11	0.48	16.7	20.1	0.0293	0.0794	0.0291	0.0911
0.99	0.50	2061	[960, 3756]	1669	[824, 2912]	0.09	0.58	15.5	22.3	0.0276	0.0619	0.0272	0.0865
0.99	0.25	2134	[918, 4138]	1584	[731, 2849]	0.07	0.52	11.0	24.0	0.0297	0.0574	0.0252	0.0857
0.975	0.50	2085	[984, 3770]	1696	[834, 2968]	0.09	0.57	15.3	22.3	0.0277	0.0619	0.0272	0.0866
0.975	0.25	2120	[932, 4020]	1592	[742, 2859]	0.07	0.52	11.1	24.0	0.0296	0.0575	0.0251	0.0858
0.965	0.75	1975	[1265, 2729]	1557	[803, 2479]	0.10	0.54	16.4	21.2	0.0285	0.0702	0.0291	0.0888
0.965	0.50	2131	[995, 3890]	1718	[851, 3002]	0.09	0.57	15.3	22.3	0.0277	0.0620	0.0272	0.0867
0.965	0.25	2196	[955, 4254]	1602	[749, 2900]	0.07	0.51	11.1	24.1	0.0298	0.0581	0.0252	0.0859

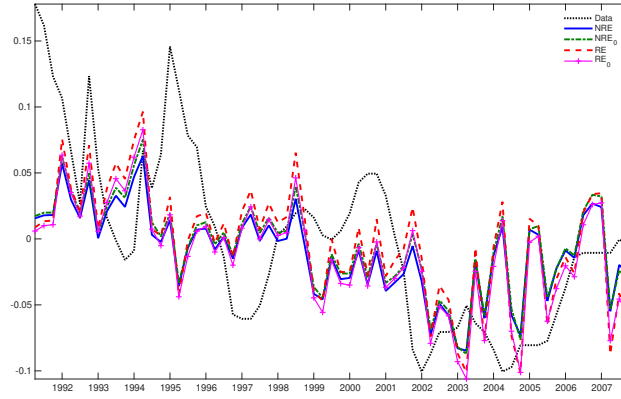
Note. Each row reports the re-estimation of the four specifications (NRE, RE, NRE₀, RE₀) at the indicated (β, ζ) combination; the first row reproduces the baseline calibration. The column labeled $\Delta\text{Log MDD}$ reports the log marginal data density of the NRE (NRE₀) specification minus that of its RE (RE₀) counterpart. The policy-rate RMSE is computed for the demeaned, annualized policy rate, as in Figure 3. Priors and the estimation procedure are identical to the baseline estimation throughout, and full posterior distributions for all grid points are included in the replication files. The posterior mean, 5th percentile, and 95th percentile were computed over one million draws from the MCMC chain.

Figure E.8: Data and Model Predictions on Monetary Policy Rates: by Extended Sample Periods



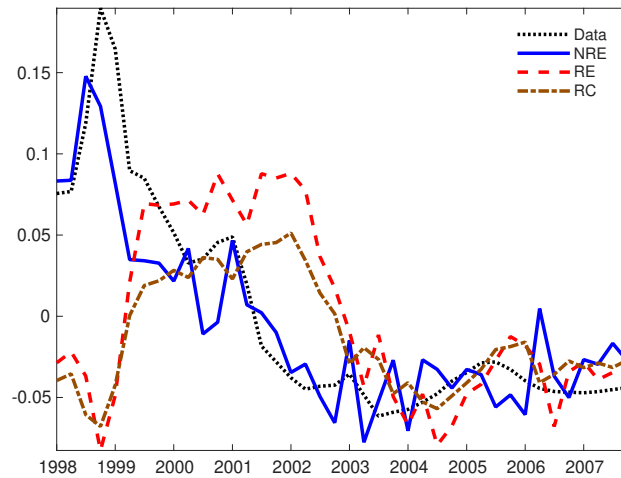
Note. The black dotted line denotes the monetary policy rate in the data. The blue solid line (labeled NRE) and the red dashed line (labeled RE) show the policy rate paths predicted by the NRE and RE models estimated over the indicated sample period, respectively. The green dash-dotted line (labeled NRE₀) and the magenta line with crosses (labeled RE₀) show the predicted policy rate paths from the estimated NRE and RE models without backward-looking firms (i.e., $\delta = 0$), respectively. All series are demeaned and annualized. In panel (a), the RMSEs for NRE, RE, NRE₀, and RE₀ are 0.0310, 0.0944, 0.0306, and 0.0860, respectively; in panel (b), they are 0.0289, 0.0574, 0.0286, and 0.0822, respectively.

Figure E.9: Data and Model Predictions on Monetary Policy Rates: Canada



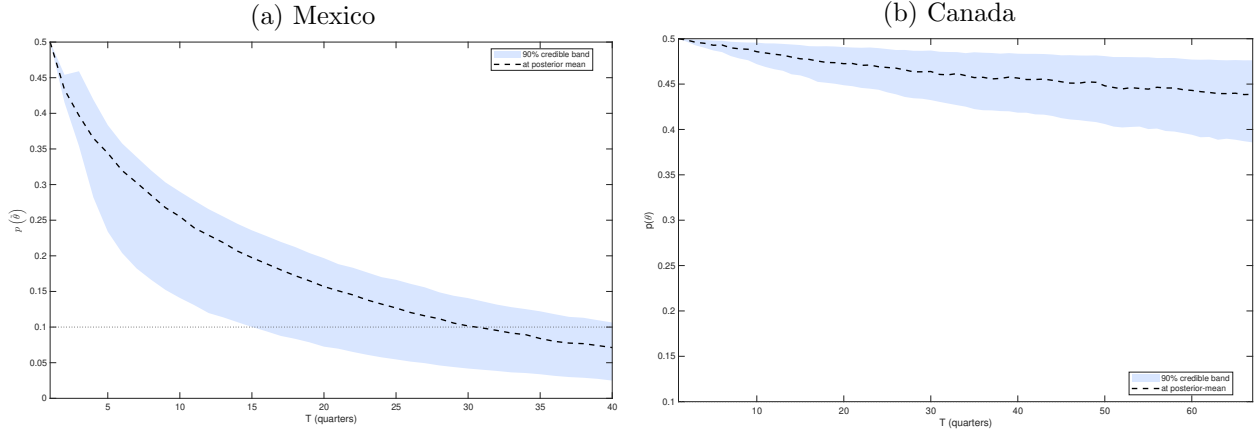
Note. The black dotted line denotes the monetary policy rate in the data. The blue solid line (labeled NRE) and the red dashed line (labeled RE) show the policy rate paths predicted by the NRE and RE models estimated on Canadian data, respectively. The green dash-dotted line (labeled NRE_0) and the magenta line with crosses (labeled RE_0) show the predicted policy rate paths from the estimated NRE and RE models without backward-looking firms (i.e., $\delta = 0$), respectively. All series are demeaned and annualized. The RMSEs for NRE, RE, NRE_0 , and RE_0 are 0.0305, 0.0398, 0.0313, and 0.0380, respectively.

Figure E.10: Model Predictions of Monetary Policy Rates in Mexico: RE Counterfactual (RC)



Note. The brown dash-dot line depicts the model-implied monetary policy rate under the RE counterfactual (“RC”), which uses the estimated NRE model parameters while setting $\hat{\theta} = 0$. The actual data on the monetary policy rate (“Data”) and the predictions from the NRE and RE models (“NRE” and “RE,” respectively) are the same as those shown in Figure 3 in Section 3.

Figure E.11: Posterior Distribution of the Detection Error Probability



Note: Each panel plots the detection error probability $p(\tilde{\theta})$ as a function of the sample length T (in quarters), computed from the posterior distribution of the baseline NRE model for each country. For each posterior draw, $p(\tilde{\theta})$ is obtained from 10,000 simulated likelihood-ratio comparisons between the worst-case model (NRE) and the approximating model (PRE) in which private agents hold rational expectations while the central bank follows the same robustly optimal rule, evaluated at that draw's parameter vector. The shaded area is the 90 percent credible band across posterior draws, and the dashed line is $p(\tilde{\theta})$ evaluated at the posterior-mean parameter estimates. The horizontal dotted line in Panel (a) marks the 0.1 reference level. The horizontal axis runs to the estimation sample size of each country ($T = 40$ for Mexico and $T = 67$ for Canada).